

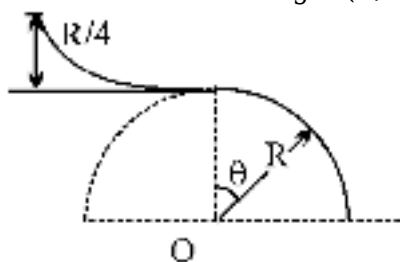
PART 1 : PHYSICS

SECTION-I

1) A man in a balloon rising vertically with an acceleration of 4.9 m/s^2 releases a ball 2 sec after the balloon is let go from the ground. The greatest height above the ground reached by the ball is ($g = 9.8 \text{ m/sec}^2$)

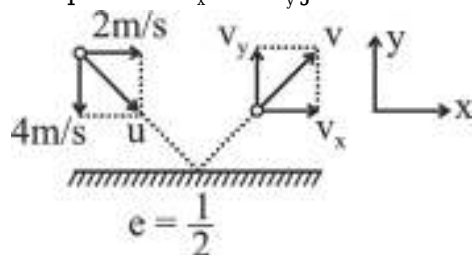
- (A) 14.7 m
- (B) 19.6 m
- (C) 9.8 m
- (D) 24.5 m

2) A skier plans to ski a smooth fixed hemisphere of radius R . He starts from rest from a curved smooth surface of height $(R/4)$. The angle θ at which he leaves the hemisphere is



- (A) $\cos^{-1} (2/3)$
- (B) $\cos^{-1} (5/\sqrt{3})$
- (C) $\cos^{-1} (5/6)$
- (D) $\cos^{-1} (5/2\sqrt{3})$

3) In the figure shown, a small ball hits obliquely a smooth and horizontal surface with speed u whose x and y components are indicated. If the coefficient of restitution is $\frac{1}{2}$, then its x and y components v_x and v_y just after collision are respectively

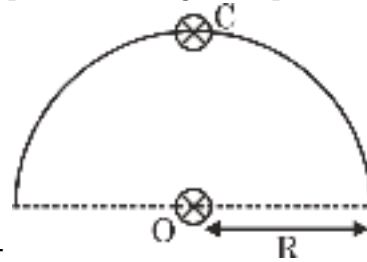


- (A) 4 m/s, 1 m/s
- (B) 2 m/s, 1 m/s
- (C) 2 m/s, 2 m/s
- (D) 4 m/s, 2 m/s

4) Seven identical geese are flying south together at constant speed. A hunter shoots one of them, which immediately dies and falls to the ground. The other six continue flying south at the original speed. After the one goose has hit the ground, the center of mass of the all seven geese :-

- (A) Continues south at the original speed, but is now located some distance behind the flying geese.
- (B) Continues south, but at $\frac{6}{7}$ the original speed.
- (C) Continues south, but at $\frac{1}{7}$ the original speed.
- (D) Stops with the dead goose.

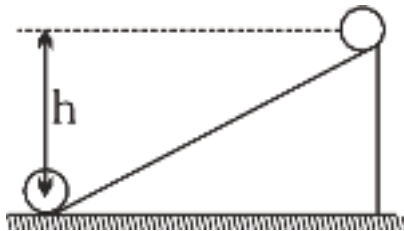
5) A semicircular ring of mass 2kg and radius of 2m is shown. The moment of inertia of this ring about the axis perpendicular to the plane of ring and passes through the points 'O' and 'C' will be I_0



and I_C respectively. Choose the most correct option :-

- (A) $I_0 = 8 \text{ kg-m}^2$
- (B) The ratio of I_C with I_0 will be 2.
- (C) $I_C - I_0 = 8$
- (D) All of the above

6) A ring of mass M and radius R is at rest at the top of an incline as shown. The ring rolls down the plane without slipping. When the ring reaches bottom, its angular momentum about its center of mass is :

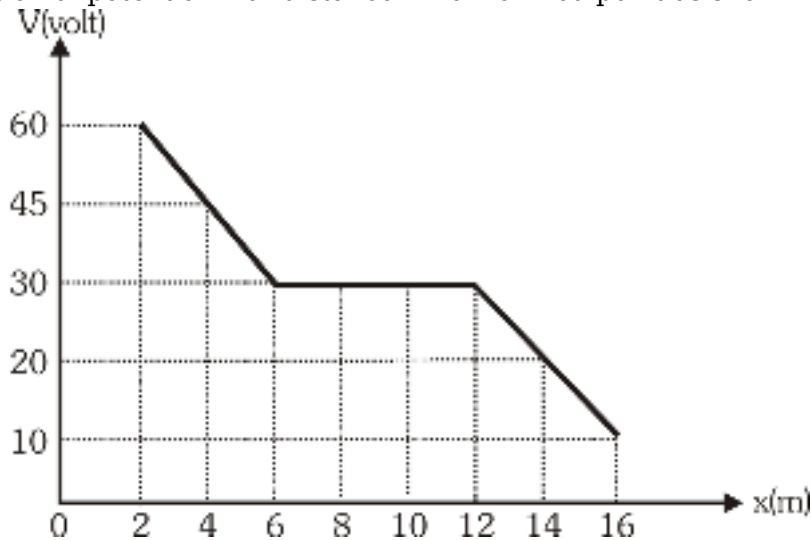


- (A) $\frac{MR}{\sqrt{gh}}$
- (B) $MR\sqrt{\frac{gh}{2}}$
- (C) $\frac{MR}{\sqrt{2gh}}$
- (D) None of these

7) A particle performs S.H.M. of amplitude A with angular frequency ω along a straight line. When it is at a distance $\frac{\sqrt{3}}{2} A$ from mean position, its kinetic energy gets increased by an amount $\frac{1}{2} m\omega^2 A^2$ due to an impulsive force. Then its new amplitude becomes-

- (A) $\frac{\sqrt{5}}{2} \text{ A}$
 (B) $\frac{\sqrt{3}}{2} \text{ A}$
 (C) $\sqrt{2} \text{ A}$
 (D) $\sqrt{5} \text{ A}$

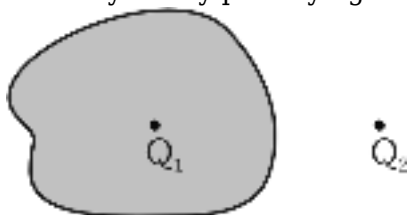
8) The variation of potential with distance x from a fixed point as shown in figure. The electric field



at $x = 13\text{m}$ is

- (A) 7.5 volt/meter
 (B) -7.5 volt/meter
 (C) 5 volt/meter
 (D) -5 volt/meter

9) Figure shows a charge Q_1 enclosed by a closed surface, while Q_2 is kept outside the closed surface. List-I gives different situation of charge & List-II gives possible value of electric field intensity at any point lying on the closed surface and total flux linked with the closed surface.



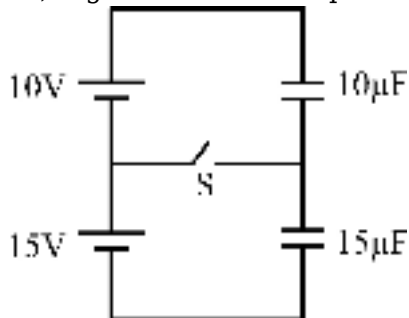
List-I		List-II	
(P)	$Q_1 = 0$ and Q_2 is positive	(1)	$E = 0, \phi = 0$
(Q)	Both Q_1 & Q_2 are positive	(2)	$E \neq 0, \phi = 0$
(R)	Q_1 is positive and Q_2 is -ve	(3)	$E \neq 0, \phi \neq 0$
(S)	Both Q_1 & Q_2 are -ve	(4)	$E = 0, \phi \neq 0$

- (A) P→2 Q→3,4 R→3,4 S→3,4
 (B) P→1 Q→1,3 R→3,4 S→3,4
 (C) P→3,4 Q→1,2 R→3 S→4
 (D) P→1 Q→2 R→3 S→4

10) A satellite of mass m is launched from earth surface to an orbit of radius $2R$, after that it is shifted from this orbit to another orbit of radius $3R$. How much energy is required to launch it from surface of earth to the orbit of radius $2R$, and to shift from this orbit to an orbit of radius $3R$. [Where R is radius of earth and g is acceleration due to gravity on earth surface]

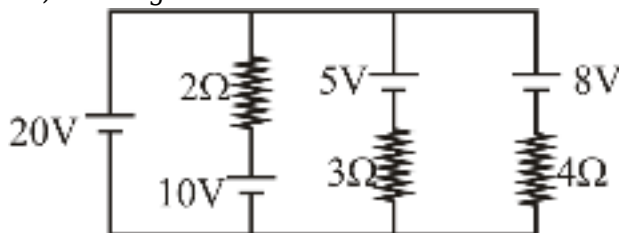
- (A) $\frac{3}{4} mgR, \frac{1}{12} mgR$
 (B) $\frac{mgR}{4}, \frac{mgR}{12}$
 (C) $\frac{3mgR}{4}, \frac{mgR}{6}$
 (D) None of these

11) Figure shows the capacitors in steady state, with S open. When S is closed :-



- (A) Both batteries supply energy & heat is lost.
 (B) Both batteries supply energy to increase the energy of both the capacitors separately
 (C) One battery supplies & other absorbs energy and no heat loss takes place
 (D) One battery supplies energy, other absorbs and heat is lost in the process.

12) In the given circuit which resistor consumes maximum power :-



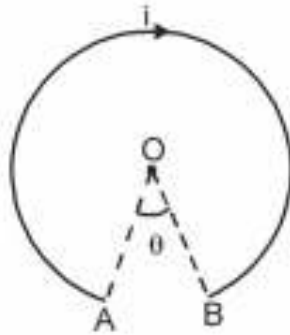
- (A) 2Ω
 (B) 3Ω
 (C) 4Ω
 (D) All consumes equal/over

13) Two straight infinitely long and thin parallel wires are spaced 0.1 m apart and carry a current of 10 ampere each. Find the magnetic field at a point which is at a distance of 0.1 m from both wires in the two cases when the currents are in the (a) same and (b) opposite directions

- (A) when currents are in same direction, $B = 2\sqrt{3} \times 10^{-8} \text{ T}$
 (B) when current are in opposite direction $B = 2 \times 10^{-5} \text{ T}$
 (C) when currents are in same direction, $B = 2 \times 10^{-5} \text{ T}$

(D) when currents are in opposite direction $B = 2\sqrt{3} \times 10^{-5} \text{T}$

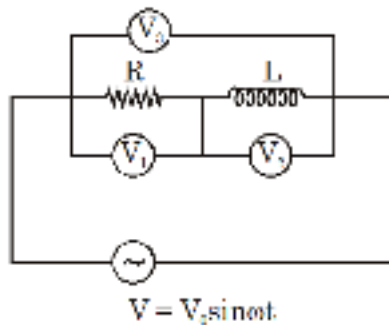
14) A current carrying wire AB of length $2\pi R$ is turned along a circle, as shown in figure. The



magnetic field at the centre O.

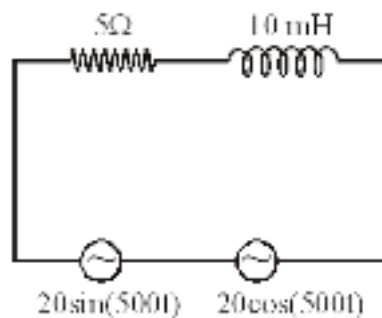
- (A) $\frac{\mu_0 i}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)^2$
- (B) $\frac{\mu_0 i}{2R} \left(\frac{2\pi - \theta}{2\pi} \right)$
- (C) $\frac{\mu_0 i}{2R} (2\pi - \theta)$
- (D) $\frac{\mu_0 i}{2R} (2\pi + \theta)^2$

15) A resistance & ideal inductor is connected in the A.C. circuit. Here V_1 , V_2 & V_3 are the reading of



three hotwire ideal voltmeters.

- (A) $V_3 = V_2 + V_1$
- (B) $V_3 > (V_1 + V_2)$
- (C) $V_3 < (V_1 + V_2)$
- (D) None



16) Current in this circuit as a function of time will be:-

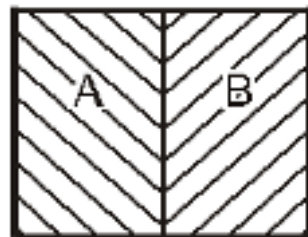
- (A) $2\sqrt{2} \sin 500t$

- (B) $4 \sin 500t$
- (C) $4 \sin(500t + 45^\circ)$
- (D) $2\sqrt{2} \sin (500t + 45^\circ)$

17) The radius of electron in a H-atom changes from $2.12 \text{ \AA}$ to $4.77 \text{ \AA}$. The H-atom has :-

- (A) Absorbed a photon of energy 1.51 eV
- (B) Absorbed a photon of energy 1.89 eV
- (C) Emitted a photon of energy 10.2 eV
- (D) Emitted a photon of energy 1.04 eV

18) A wall is made of two layers A and B of the same thickness but different materials as shown. The thermal conductivity of A is thrice that of B. In steady state, the temperature difference across the

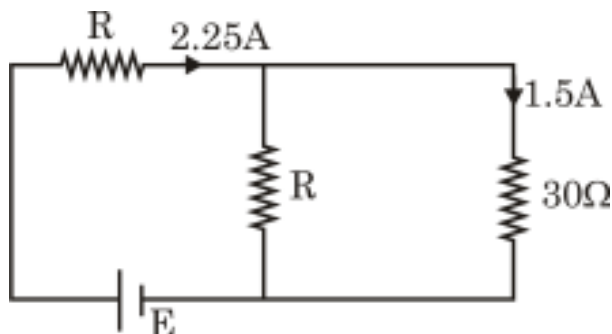


wall is 36°C . The temperature difference across the layer A is :

- (A) 6° C
- (B) 9° C
- (C) 18° C
- (D) 27° C

19) If $n \gg 1$ then the dependence of frequency of photon emitted as a result of transition of an electron from n^{th} orbit to $(n - 1)^{\text{th}}$ orbit in the hydrogen atom on n will be

- (A) $\nu \propto \frac{1}{n}$
- (B) $\nu \propto \frac{1}{n^2}$
- (C) $\nu \propto \frac{1}{n^3}$
- (D) $\nu \propto \frac{1}{n^4}$



20) In the circuit shown, find the value of E/R .

- (A) 3
- (B) 5
- (C) 7
- (D) 8

SECTION-II

1) A coil of area 3.2 cm^2 made of 50 turns of thin wire connected to a ballistic galvanometer of 200 ohm resistance with a constant of $2 \times 10^{-5} \text{ C/div}$ is inserted a between pole pieces of electromagnet. When the coil is withdrawn from the field, the galwonometer pointer moves 20 divisions. What is the magnetic field (in T). (The ballistic galvanometer measures the charge flown through it).

2) A 1-km steel rail track was laid out (when the ambient temperature was 20°C) without any gaps for expansion. When the ambient temperature rose to 25°C , the rail track buckled into an isosceles triangle. If the coefficient of linear expansion of steel is $14 \times 10^{-6} \text{ K}^{-1}$, what is the height of the buckle (in m) (to the nearest integer)?

3) A cylindrical bucket of depth 60 cm is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as seen by the observer is given as $\alpha \text{ cm}$. Fill $\left(\frac{\alpha}{5}\right)$ in OMR sheet.

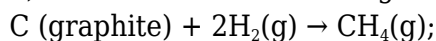
4) A flood light is covered with a filter that transmits red light. The electric field of the emerging beam is represented by a sinusoidal plane wave $E_x = 36 \sin (1.20 \times 10^7 z - 3.6 \times 10^{15} t) \text{ V/m}$. The average intensity of the beam is close to $\sqrt{n}$, where n is integer. Find n.

5) A horizontal stretched string, fixed at two ends, is vibrating in its 39th overtone according to the equation, $y(x, t) = (0.01\text{m}) \sin [(62.8 \text{ m}^{-1})x] \cos [(628 \text{ s}^{-1})t]$. Assuming $\pi = 3.14$, what is the length of string (in m) ?

PART 2 : CHEMISTRY

SECTION-I

1) Calculate ΔS for following chemical reaction



The standard entropies of C (graphite), $\text{H}_2(\text{g})$ and $\text{CH}_4(\text{g})$ are 5.740, 130.684 and 186.264 J/K-mol, respectively.

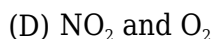
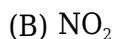
(A) $-80.844 \frac{\text{J}}{\text{K}}$

(B) $125 \frac{\text{J}}{\text{K}}$

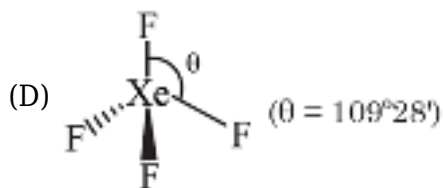
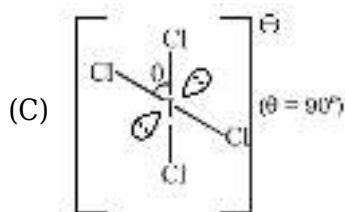
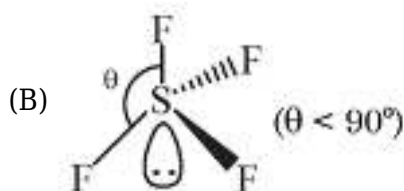
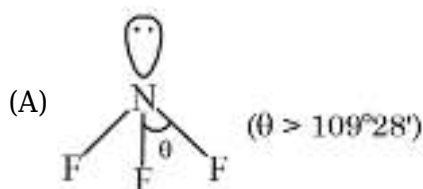
(C) $212 \frac{\text{J}}{\text{K}}$

(D) $138.2 \frac{\text{J}}{\text{K}}$

2) Some white colourless crystals are heated. A cracking sound is heard and brown fumes are given off and the residue left is yellow-brown in colour. When a glowing splinter is held in the fumes, it is relighted. The fumes consist of :



3) Which of the following structure is correctly drawn according to fundamental idea of **VSEPR** theory-



4) A proton accelerated from rest through a potential difference of 'V' volts has a wavelength λ associated with it. An alpha particle in order to have the same wavelength must be accelerated from

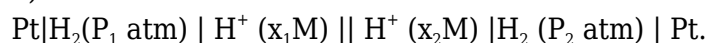
rest through a potential difference of

- (A) V volt
- (B) 4V volt
- (C) 2V volt
- (D) $\frac{V}{8}$ volt

5) On reacting with KMnO_4 , FeSO_4 converts into Fe^{+3} in acidic medium. If 250 ml of FeSO_4 reacts with 32 ml of 5 M solution of KMnO_4 . The molarity of FeSO_4 solution is

- (A) 3.2 M
- (B) 1 M
- (C) $\frac{4}{3}$ M
- (D) $\frac{2}{3}$ M

6) Consider the cell :



The cell reaction will be spontaneous if -

- (A) $\text{P}_1 = \text{P}_2$ and $\text{x}_1 > \text{x}_2$
- (B) $\text{P}_1 = \text{P}_2$ and $\text{x}_1 = \text{x}_2$
- (C) $\text{x}_1 = \text{x}_2$ and $\text{P}_1 > \text{P}_2$
- (D) $\text{x}_1 = \text{x}_2$ and $\text{P}_1 < \text{P}_2$

7) Upon introduction of a catalyst at 500 K, the rate of a first order reaction increases to 2.718 times. If the activation energy in the presence of a catalyst is 4.15 kJ mol^{-1} . Then the E_a in absence of catalyst is ($R = 8.3 \text{ J/K. mole}$ $\ln 2.718 = 1$)

- (A) 4.15 kJ
- (B) 2.08 kJ
- (C) 2.718 kJ
- (D) 8.3 kJ

8) Geometry, hybridization, and magnetic moment of the ions

$[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{MnBr}_4]^{2-}$ and $[\text{FeF}_6]^{4-}$, respectively are:

- (A) Tetrahedral, square planar, octahedral: sp^3 , dsp^2 , sp^3d^2 : 5.9, 0, 4.9
- (B) Tetrahedral, square planar, octahedral: dsp^2 , sp^3 , sp^3d^2 : 0, 5.9, 4.9
- (C) Square planar, tetrahedral, octahedral: dsp^2 , sp^3 , d^2sp^3 : 5.9, 4.9, 0 .
- (D) Square planar, tetrahedral, octahedral: dsp^2 , sp^3 , sp^3d^2 : 0, 5.9, 4.9

9) Potential energy of electron present in 2^{nd} orbit of Li^{2+} is : (r_0 = Radius of 1st Bohr's orbit of hydrogen)

- (A) $\frac{e^2}{4\pi\epsilon_0 r_0}$
- (B) $-\frac{3e^2}{4\pi\epsilon_0 r_0}$
- (C) $-\frac{3e^2}{16\pi\epsilon_0 r_0}$
- (D) $-\frac{9e^2}{16\pi\epsilon_0 r_0}$

10) Consider the following statements. Select them as "T" (true) or "F" (false) :

- (I) Ionization energy for s-electron is more than p-electrons of the same shell since s-electrons are closest to the nucleus
- (II) EA of oxygen is less than that of fluorine but greater than that of nitrogen
- (III) IE of Mg is more than that of aluminium
- (IV) Bond energy of F_2 is less than Cl_2

- (A) T T T T
- (B) F F F F
- (C) T F F F
- (D) T T T F

11) According to Molecular orbital theory which of the following is correct ?

- (A) Bond order of N_2^- is 2.5
- (B) In C_2 molecules both the bonds are π bonds
- (C) In C_2^{2-} - ion there is one σ and two π bonds
- (D) All the above are correct

12) The value of enthalpy change (ΔH) for the reaction $C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$ is $-1366.5 \text{ kJ mol}^{-1}$. The value of internal energy change for the above reaction at 27°C this temperature will be:

- (A) -1371.5 kJ
- (B) -1369.0 kJ
- (C) -1364.0 kJ
- (D) -1361.5 kJ

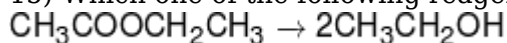
13) Which of the following order is correct for thermal stability ?

- (A) $K_2CO_3 < MgCO_3 < CaCO_3 < BeCO_3$
- (B) $BeCO_3 < MgCO_3 < CaCO_3 < K_2CO_3$
- (C) $MgCO_3 < BeCO_3 < K_2CO_3 < CaCO_3$
- (D) $CaCO_3 < MgCO_3 < BeCO_3 < K_2CO_3$

14) Which of the following is correct ?

- (A) D(+)-glucose and D(+)-mannose are enantiomers.
 (B) D(+)-glucose and D(-)-fructose give same osazone.
 (C) D(-)-fructose give single product on reduction with Na-Hg.
 (D) D(+) -glucose and D(-) -fructose are epimers.

15) Which one of the following reagent can be used for given conversion?



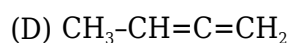
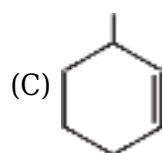
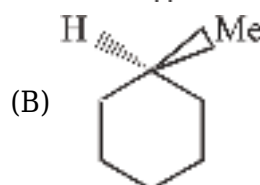
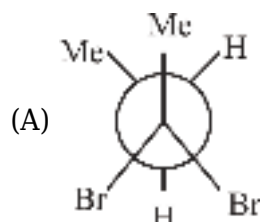
(only organic product)

- (A) $\text{NaBH}_4/\text{EtOH}$
 (B) (i) $\text{LiAlH}_4/\text{ether}$. (ii) H_3O^+
 (C) (i) CH_3MgBr (ii) H_3O^+
 (D) I_2/NaOH

16) Which of the following reagent/series of reagents can be used for distinction between methyl alcohol and ethyl alcohol?

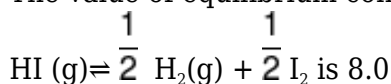
- (A) Conc. HCl/ZnCl_2
 (B) $\text{CH}_3\text{COOH}/\text{H}^+$
 (C) NaOH/I_2
 (D) None of these

17) Which of the following compound can exhibit optical isomerism?

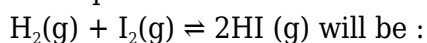


18)

The value of equilibrium constant of the reaction



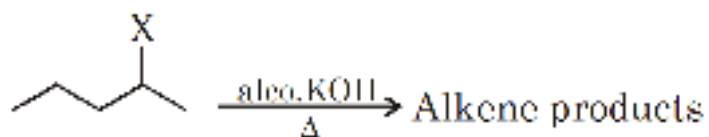
The equilibrium constant of the reaction



will be :

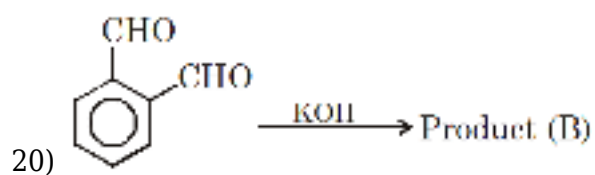
- (A) $\frac{1}{64}$
 (B) 16
 (C) $\frac{1}{8}$
 (D) $\frac{1}{16}$

19) In the reaction



Maximum extent of hofmann alkene product will be obtained when X is-

- (A) I
 (B) Br
 (C) Cl
 (D) F



Product 'B' in the above reaction is-

- (A)
- (B)
- (C)
- (D)

SECTION-II

1) How many of the following gives flame test :

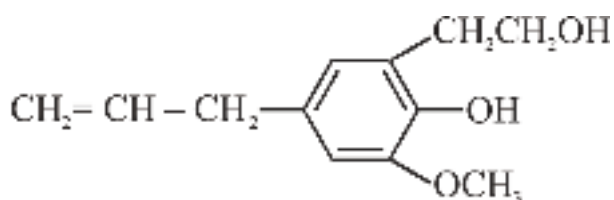
Li, Na, K, Be, Mg, Ca, Ba, Cs

2) Among the following, how many reagents are used to give black precipitate of metal sulphide ?

AgNO_3 , BaCl_2 , $\text{Pb}(\text{OAc})_2$, HgCl_2 , CuSO_4 , SbCl_3 .

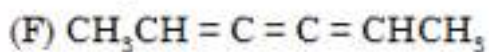
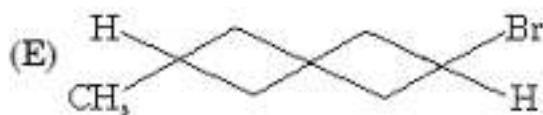
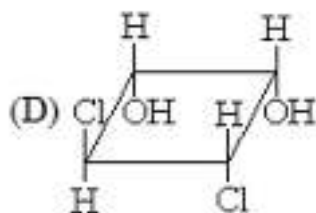
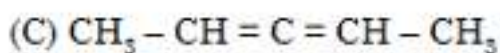
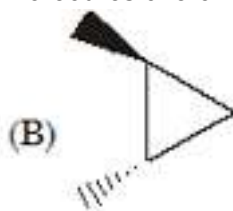
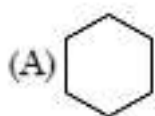
3)

Calculate number of moles of HI required to react with one mole of the given compound -



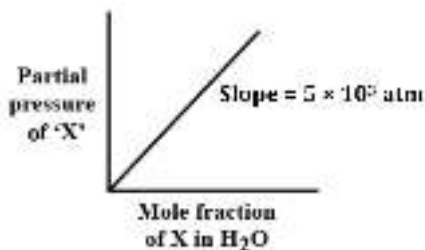
4)

How many of the following molecules are chiral ?



5) A gas 'X' is present with saturated water vapour over water liquid at total pressure of 1.5 atm.

Vapour pressure of H_2O at same temperature is 0.5 atm. If number of mole of gas dissolved in 10 mole of water is $x \times 10^{-3}$ then calculate x.



PART 3 : MATHEMATICS

SECTION-I

1) If $a + b + c = 0$, then the value of expression $a^6 + b^6 + c^6 - \frac{(a^2 + b^2 + c^2)^3}{4} - \frac{(a^3 + b^3 + c^3)^2}{3}$ is -

- (A) negative for some values of a, b & c
- (B) positive for some negative values of a, b & c
- (C) independent of values of a, b & c
- (D) independent of values of a, b & c only if $a, b, c \in \mathbb{R}^+$

2) The sum of absolute maximum and absolute minimum values of the function

$f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ in the interval $[0, 1]$ is:

- (A) $3 + \frac{\sin(1) \cos^2\left(\frac{1}{2}\right)}{2}$
- (B) $3 + \frac{1}{2}(1 + 2 \cos(1)) \sin(1)$
- (C) $5 + \frac{1}{2}(\sin(1) + \sin(2))$
- (D) $2 + \sin\left(\frac{1}{2}\right) \cos\left(\frac{1}{2}\right)$

3) The set of point where $f(x) = \frac{x}{1 + |x|}$ is differentiable is

- (A) $(-\infty, -1) \cup (-1, \infty)$
- (B) $(-\infty, \infty)$
- (C) $(0, \infty)$
- (D) $(-\infty, 0) \cup (0, \infty)$

4) Solution of the equation

$$\cos(3^x \pi) - 2 \cos^2(3^x \pi) + 2 \cos(4(3^x) \pi) - \cos(7(3^x) \pi)$$

$$= \sin(3^x \pi) + 2 \sin^2(3^x \pi) - 2 \sin(4(3^x) \pi) + 2 \sin(3^{x-1} \pi) - \sin(7(3^x) \pi)$$

is

(A) $x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{1}{16} - \frac{1}{16} \right), n \in \mathbb{Z}$

(B) $x(2k+1)\pi - \frac{\pi}{5} \cup x = 2m\pi; m \in \mathbb{I}, k \in \mathbb{I}$

(C) $x = (2k+1)\pi - \frac{\pi}{3} \cup x = 2m\pi, m \in \mathbb{I}, k \in \mathbb{I}$

(D) $x = (2k+1)\pi - \frac{\pi}{2} \cup x = 2m\pi, m \in \mathbb{I}, k \in \mathbb{I}$

$$f(x) = \frac{x^2 - |x| - 2}{2} + \left| \frac{x^2 + |x| - 6}{2} \right|$$

5) Let function $f(x)$ be the number of points where $f(x)$ is non differentiable and m be the number of real solutions of equation $f(x) = 2$ then the value of $(\ell + m)$ is :

(A) 4

(B) 5

(C) 6

(D) 3

6) Let the set of complex numbers $(a_1, b_1); (a_2, b_2), (a_3, b_3) \dots$ denoting the points on the complex plane satisfying $(a_{n+1}, b_{n+1}) = (\sqrt{3}a_n - b_n, \sqrt{3}b_n + a_n)$ for $n = 1, 2, 3, \dots$. Suppose $(a_{100}, b_{100}) = (2, 4)$ then the value of $(a_1 + b_1)$ is equal to

(A) $\frac{1}{2^{96}}$

(B) $\frac{1}{2^{97}}$

(C) $\frac{1}{2^{98}}$

(D) $\frac{1}{2^{99}}$

7) Let P_1 be a regular polygon of n sides with side length x and let P_{k+1} be the regular polygon of n sides formed by joining the mid points of consecutive sides of P_k . Also perimeters of the P_i 's form an infinite geometric progression. If $n = 6$ and $x = \frac{1}{2 + \sqrt{3}}$, then the sum of this infinite geometric progression is :

(A) 8

(B) 17

(C) 14

(D) 12

$$e^x \left(\int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 \right) = 1$$

8) Let $f(x)$ be a differentiable function satisfying Then the

value of $f(2)$ is

- (A) e^2
- (B) $\frac{1}{e^2}$
- (C) $e + e^2$
- (D) $2e^2$

$$\int_1^e \frac{f(t)}{t} dt$$

9) Let $f : (0, \infty) \rightarrow \mathbb{R}$ be continuous function such that $f(x) = 6 (\ln x)^2 - 1$.

Let Δ be area of triangle formed by tangent at $x = \frac{1}{e}$ to curve $y = f(x) + 1$ and co-ordinate axes. Then the value of $\left\lfloor \frac{2e\Delta}{3} \right\rfloor$ is equal to _____. [where $\lfloor . \rfloor$ represents G.I.F.]

- (A) 3
- (B) 4
- (C) 6
- (D) 9

10) If $f(x) = \int \frac{(e^{2x} - e^x + 1)dx}{(e^x \sin x + \cos x)(e^x \cos x - \sin x)}$. If $f(0) = 2025$, then $f(2024\pi)$ is equal to

- (A) $-2024\pi + 2025$
- (B) $2024\pi - 2025$
- (C) $-2025\pi + 2024$
- (D) None of these

11) Let P_1 and P_2 be two parabolas with distinct directrices ℓ_1 and ℓ_2 and distinct foci F_1 and F_2 respectively. It is known that $F_1 F_2$ is parallel to ℓ_1 and ℓ_1 and ℓ_2 are parallel. F_1 lies on P_2 and F_2 lies on P_1 . The two parabolas intersect at distinct points A and B. Given that $F_1 F_2 = 1$, the value of AB^2 can be

- (A) 2
- (B) 3
- (C) 4
- (D) $\frac{15}{4}$

12) For a bijective function $f: \mathbb{R} \rightarrow \mathbb{R}$ we say $g: \mathbb{R} \rightarrow \mathbb{R}$ is ALLENITE if it satisfies the identity $(f \circ g)(x) = g^{-1}(x)$ where $g^{-1}(x)$ is inverse function of $g(x)$. Given $g(x) = x^3 + 9x^2 + 27x + 81$ and f is its ALLENITE then $|f(-289)|$ is

- (A) 5
- (B) 6
- (C) 7

(D) 8

13) Let a hyperbola $H : \frac{x^2}{3} - y^2 = 1$ which is orthogonal to an ellipse E , whose major and minor axis are respectively transverse and conjugate axis of H . Centres of H and E coincide and both are confocal product of their eccentricities is 1. If tangent and normal of H at point P (where P is point of intersection of H & E) intersects its transverse axis at T_1 and N_1 respectively then radius of incircle of triangle T_1PN_1 is

(A) $\frac{\sqrt{108}}{\sqrt{84} + \sqrt{63} + \sqrt{36}}$

(B) $\frac{1}{\frac{1}{\sqrt{84}} + \frac{1}{\sqrt{63}} + \frac{1}{\sqrt{36}}}$

(C) $\frac{\frac{1}{\sqrt{108}}}{\frac{1}{\sqrt{84}} + \frac{1}{\sqrt{63}} + \frac{1}{\sqrt{36}}}$

(D) $\frac{1}{\sqrt{84}} + \frac{1}{\sqrt{63}} + \frac{1}{\sqrt{36}}$

14) An isoscles triangle ABC ($AB = AC$) is inscribed in circle $x^2 + y^2 = 16$. $A(0,4)$. From point A, B and

C three ordinates are drawn which cuts the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the points P, Q and R respectively such that A and P are on opposite side; Q and B are on the same side; C and R are on same side of major axis. If ΔPQR is right angled at P and ϕ is the smallest angle of ΔABC , then-

(A) $\tan \phi = \frac{3}{4}$

(B) $\tan \phi = \frac{7}{24}$

(C) coordinates of point R is $\left(\frac{96}{25}, \frac{28}{25}\right)$

(D) coordinates of point R is $\left(\frac{28}{25}, \frac{96}{25}\right)$

15) Let n be the number of integral values of x , which satisfy the equation

$$\tan^{-1}x - \sin^{-1}x = \cos^{-1}x - \cot^{-1}x.$$

Let $\vec{c} = \left| \vec{a} \times \vec{b} \right| \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b}$ where $\vec{a}$ and $\vec{b}$ are unit vectors and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$ then the value of $(n + 2 |\vec{c}|^2)$ is equal to

(A) 10

(B) $\frac{17}{2}$

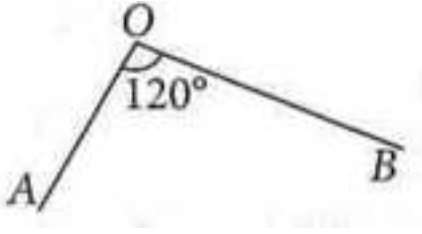
(C) $\frac{13}{2}$

(D) 30

16) Let $f(x)$ be differentiable at $x = 0$. If the normal to the curve $y = f(x)$ at $x = 0$ be given by the equation $3x - y + 3 = 0$ then the value of $\lim_{x \rightarrow 0} x^2 \{ f(x^2) - 5f(4x^2) + 4f(7x^2) \}^{-1}$ is equal to

- (A) $-\frac{1}{5}$
 (B) $-\frac{1}{4}$
 (C) $-\frac{1}{3}$
 (D) $-\frac{1}{2}$

17) OA and OB are two roads enclosing an angle of 120° . X and Y start from 'O' at the same time. X travels along OA with a speed of 4 km/hour and Y travels along OB with a speed of 3 km/hour. The rate at which the shortest distance between X and Y is increasing after 1 hour is



- (A) $\sqrt{37}$ km/h
 (B) 37 km/h
 (C) 13 km/h
 (D) $\sqrt{13}$ km/h

18) Consider $f(x) = \begin{cases} 2 - |x| & -1 \leq x \leq 1 \\ |x - 2| - x & 1 < x \leq 3 \end{cases}$ and $g(x) = \begin{cases} \sin x - 1 & 0 \leq x < \frac{\pi}{2} \\ [x] - \cos(x - 2) & \frac{\pi}{2} \leq x \leq \pi \end{cases}$

(where $[.]$ represents greatest integer function) then which of following is correct.

- (A) $\lim_{x \rightarrow 1^+} g(f(x)) = -1$
 (B) $\lim_{x \rightarrow 0^-} g(f(x)) = 0$
 (C) $\lim_{x \rightarrow 2^+} \frac{f(g(x))}{f(x) - 2} = \frac{1}{2}$
 (D) $\lim_{x \rightarrow 0^+} \frac{g(f(x))}{(f(x) - 2)^2} = \frac{-1}{2}$

19) Let E and F be two independent events. The probability that both E and F happen is $\frac{1}{12}$ and the probability that neither E nor F happens is $\frac{1}{2}$, then a value of $\frac{P(E)}{P(F)}$ is :

- (A) $\frac{4}{3}$
 (B) $\frac{1}{3}$
 (C) $\frac{5}{12}$
 (D) $\frac{3}{2}$

20) Consider the points $A(\vec{0})$, $B(3\hat{i} + f(t)\hat{j} + \hat{k})$ and $C(2\hat{i} + f'(t)\hat{j} + 2\hat{k})$ and another point D are coplanar for all $t \in \mathbb{R}$

- (A) if D is $(\hat{i})$ then $f(x) = 2 + c \ln(2x)$, $c \in \mathbb{R}$
 (B) if D is $(\hat{i})$ then $f(x) = c \ln(2x)$, $c \in \mathbb{R}$
 (C) if D is $(\hat{i} + \hat{j})$ then $f(x) = 2 + ce^{2x}$, $x \in \mathbb{R}$
 (D) if D is $(\hat{i} + \hat{j})$ then $f(x) = 2 + c \ln(2x)$, $c \in \mathbb{R}$

SECTION-II

1) Let $f : [-5, 5] \rightarrow [3, 18]$ is a surjective differentiable function and $g(x) = f(x) f'(x)$ is twice differentiable function. given that $g(0) = k$, then the least integral value of k such that the always exist some $c \in (-5, 5)$ for all functions to satisfy $(f'(c))^2 = -f(c)f''(c)$ and $(f(c))^2 f'''(c) < 3(f'(c))^3$.

2) Let $I_{(m,n)} = \int (\cos 2x)^m (1 - \tan^2 x)^n dx$, such that
 $4I_{(0,10)} + \frac{I_{(1,10)}}{5} = A \sin 2x (1 - \tan^2 x)^{10} + B I_{(0,9)} + C$ (where C is integration constant), then value
 $\frac{B}{8A}$ is

3) Consider the ellipse $S_1 = \frac{x^2}{9} + \frac{y^2}{x} - 1 = 0$. Let $S_2 = 0$ be a parabola on the right of y-axis confocal with $S_1 = 0$ having vertex at the centre of $S_1 = 0$. Let P be the point of intersection of the parabola and directrix of the ellipse in the 1st quadrant. If tangent at point Q in 1st quadrant on $S_1 = 0$ and the line joining point P with the focus of $S_2 = 0$ intersect on the auxillary circle of $S_1 = 0$ then the value of $9\sin^2\theta$ is _____. (θ is the eccentric angle for the point Q).

4) If $a + b + c = 0$ and $a^2 + b^2 + c^2 = 1$, then the value of $(ab)^2 + (bc)^2 + (ac)^2$ is $\frac{p}{q}$ where p, q (p, q $\in \mathbb{N}$) are relatively prime numbers then the value of $p + q$ is

5) Let $S = \{1, 2, \dots, 2005\}$. If there is at least one prime number in any subset of S consisting of n pairwise coprime numbers, then the minimum value of n is

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	C	C	B	A	A	C	C	A	B	D	B	B	A	C	B	B	B	C	A

SECTION-II

Q.	21	22	23	24	25
A.	5	6	8	3	2

PART 2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	A	D	C	D	A	C	D	D	D	A	D	C	B	B	B	C	C	A	D	B

SECTION-II

Q.	46	47	48	49	50
A.	6	4	3	4	2

PART 3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	C	B	B	A	C	C	D	B	D	A	D	C	C	A	C	C	A	B	A	C

SECTION-II

Q.	71	72	73	74	75
A.	55	5	3	5	16

SOLUTIONS

PART 1 : PHYSICS

1)

Explain Question: Question asks maximum height reached by stone under gravity

Concept: $2g h_{\max} = v_0^2 - u^2 \Rightarrow h_{\max} = \frac{u^2}{2g}$

Solution:

$$v = 0 + 4.9 \times 2 = 9.8 \text{ m/s}$$

$$h = \frac{9.8 \times 9.8}{2 \times 9.8} = 4.9 \text{ m}$$

$$s = \frac{1}{2} \times 4.9 \times 4 = 9.8$$

Total height = 4.9 + 9.8 = 14.7 m **Final Answer:** Option (A) is Correct.

2)

Explain Question: If is asked about the position when block leaves the contact.

Concept: At leaving point normal becomes 0.

Solution:

Conserving energy between points A & B,

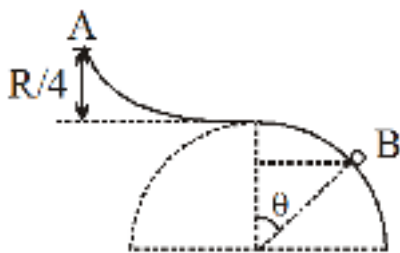
$$mg \left[\frac{R}{4} + R(1 - \cos \theta) \right] = \frac{1}{2} mv^2$$

Also EFD at point B will give,

$$\therefore \frac{1}{4} + (1 - \cos \theta) = \frac{1}{2} \cos \theta$$

$$\frac{5}{4} = \frac{3}{2} \cos \theta$$

$$\therefore \cos \theta = \frac{5}{6}$$



$$mg \cos \theta = \frac{mv^2}{R}$$

Final Answer: Option (C) is Correct.

3)

Explain Question: Velocity after collision has been asked.

Concept: After collision $V = eV_0$ (Along line of impact).

Solution:

$$v_x = \text{constant}, v_y = e \times 4 = \frac{1}{2} \times 4 = 2 \text{ m/s}$$

Final Answer: Option (C) is Correct.

4)

Explain Question: It has been asked to find velocity of CM.

$$V_{cm} = \frac{m_1 v_1 + m_2 v_2 + \dots}{m_1 + m_2 + \dots}$$

Concept:

Solution:

$$V_{cm} = \frac{6mv_0}{7m} = \frac{6v_0}{7}, \text{ direction same.}$$

Final Answer: Option (B) is Correct.

5)

Explain Question: Moment of inertia about different axis has been asked

Concept: Parallel axis theorem $I = I_{cm} + md^2$

Solution:

$$I_0 = mR^2 = 2 \times 2 \times 2 = 8\text{kgm}^2$$

We can't apply parallel theorem between I_c and I_0 because no one axis is pass through centre of mass.

Final Answer: Option (A) is Correct.

6)

Explain Question: Total angular momentum has been asked in case of pure rolling about center of mass

Concept: $L_{cm} = L\omega$

Solution:

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}(MR^2)\left(\frac{v}{R}\right)^2 \Rightarrow v = \sqrt{gh}$$

Angular momentum $L = I\omega$

$$= MR^2 \frac{\sqrt{gh}}{R} = MR \sqrt{gh}$$

Final Answer: Option (A) is Correct.

7)

Explain Question: It has been asked to find new amplitude of oscillation when same extra energy is supplied.

Concept: $E_{total} = \frac{1}{2}kA^2$

Solution:

Total energy of SHM (After impulse)

$$m\omega^2 A^2 = \frac{1}{2}m\omega^2 A_1^2$$

$$\therefore A_1 = \sqrt{2}A$$

Final Answer: Option (C) is Correct.

8)

Explain Question: The value of electric field has been asked from V-x graph.

Concept: $E = -\frac{dv}{dx} =$ slope of graph.

Solution:

$$E = -\frac{dV}{dx} = \frac{20}{4} = 5 \text{ volt/meter}$$

Final Answer: Option (C) is Correct.

9)

Explain Question: Electric flux have been asked in different situation.

Concept: Gauss law ; $\phi = \frac{q_{in}}{\epsilon_0}$

Solution:

$\phi_{net} = \frac{q_{in}}{\epsilon_0}$ field on the boundary of the closed surface can be zero as both charges contribute to the net field.

Final Answer: Option (A) is Correct.

10) **Explain the Question :**

A satellite of mass m is taken from Earth's surface to an orbit of radius 2R, then from radius 2R to radius 3R; find the energy required in each step.

Concept :

This question is based on gravitational potential energy of satellite orbitals and total mechanical energy $E = -GMm/(2r)$.

Solution :

For a satellite in circular orbit of radius r, total energy is

$$E = -GMm/(2r)$$

Using $gR^2 = GM$, we rewrite $GMm = mgR^2$

Energy at Earth's surface ($r = R$) :

$$E_1 = -(mgR^2)/(2R) = -(1/2) mgR$$

Energy at orbit $r = 2R$:

$$E_2 = -(mgR^2)/(4R) = -(1/4) mgR$$

Energy needed to go from surface to 2R :

$$\Delta E_1 = E_2 - E_1 = [-(1/4) mgR] - [-(1/2) mgR]$$

$$\Delta E_1 = (1/4) mgR$$

Now for $r = 3R$:

$$E_3 = -(mgR^2) / (6R) = - (1/6) mgR$$

Energy needed to go from 2R to 3R :

$$\Delta E_2 = E_3 - E_2 = [-(1/6) mgR] - [-(1/4) mgR]$$

$$\Delta E_2 = (1/12) mgR$$

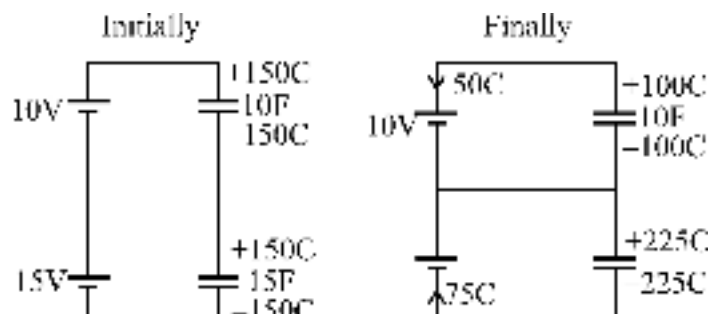
Final Answer: The correct option is (2).

11)

Explain Question: Energy supplied and lost by the batteries has been asked after switching circuits.

Concept: $\text{Loss} = W_b - \{U_f - U_i\}$

Solution:



Hence work done by one battery will be + and other will be - & since
(work by battery 1) + (work by battery 2) + $\Delta U \neq 0$

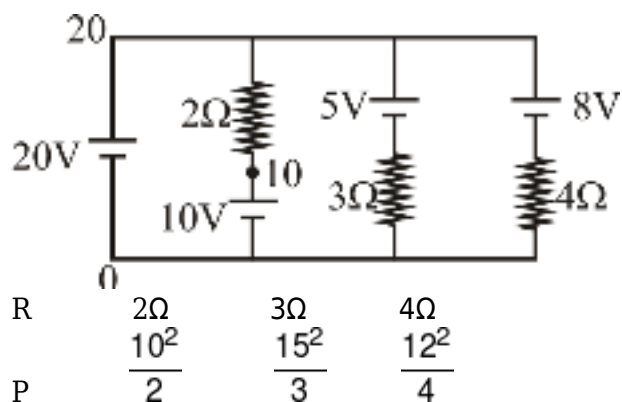
Final Answer: Option (D) is Correct.

12)

Explain Question: Maximum power consumption has been asked in a circuit.

Concept: $P = i^2 R$

Solution:



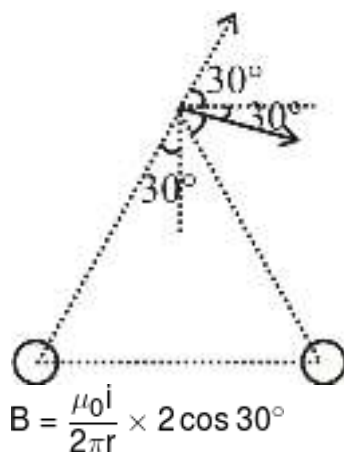
Final Answer: Option (B) is Correct.

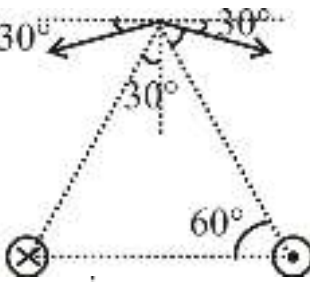
13)

Explain Question: The value of magnetic field due to two long wires has been asked

Concept: $B = \frac{\mu_0 i}{2\pi r}$

Solution:



$$= 2 \times 10^{-6} \times 10 \times \sqrt{3}$$


$$B = \frac{\mu_0 i}{2\pi r} \times 2 \cos 60^\circ$$

$$= 2 \times 10^{-5}$$

Final Answer: Option (B) is Correct.

14)

Explain Question: The value of magnetic field at the center of the arc of wire has been asked.

Concept: $B = \frac{\mu_0 i \alpha}{4\pi r}$

Solution:

$$B = \frac{\mu_0 i \alpha}{4\pi r}$$

$$\alpha = 2\pi - \theta$$

$$(2\alpha - \theta)r = 2\pi R \Rightarrow r = \frac{2\pi R}{2\pi - \theta}$$

Final Answer: Option (A) is Correct.

15)

Explain Question: Reading of voltmeter in ac circuit has been asked.

Concept: $V_{\text{net}} = \sqrt{V_1^2 + V_2^2}$

Solution:

$$V_3 = \sqrt{V_1^2 + V_2^2}$$

$$V_3^2 = V_1^2 + V_2^2$$

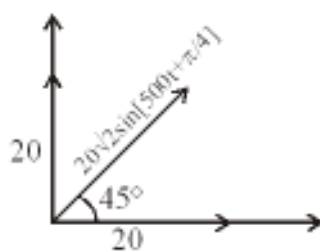
$$V_3^2 < V_1^2 + V_2^2 + 2V_1V_2$$

$$V_3 < V_1 + V_2$$

Final Answer: Option (C) is Correct.

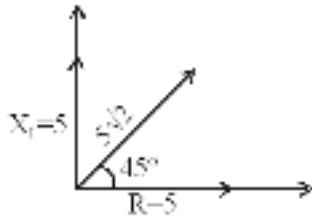
16) **Explain Question:** In the given question on LR AC circuit is given and current is asked.

Concept: R—L circuit



Solution:

Phaser diagram for the emf $20 \sin(500t)$ and $20 \cos(500t)$



Impedance diagram for the circuit $X_L = \omega_L$

$$= 500 \times 10 \times 10^{-3} = 5\Omega = \Omega$$

Both impedance and voltage are in same phase

$$i = 4 \sin 500 t$$

17)

Explain Question: Transition of electron in a H-atom has been asked about.

$$\Delta E = 13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Concept:

Solution:

$$R = n^2 a_0$$

$$2.12 = n_1^2 \times 0.53$$

$$n_1 = 2$$

$$4.77 = n_2^2 \times 0.53$$

$$n_2 = 3$$

$$E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{13.6 \times 5}{36} = \frac{68}{36}$$

Final Answer: Option (B) is Correct.

18)

Explain Question: Temperature difference across a given layer of conductor has been asked.

Concept: $\Delta T = i_{th} R_{th}$

Solution:

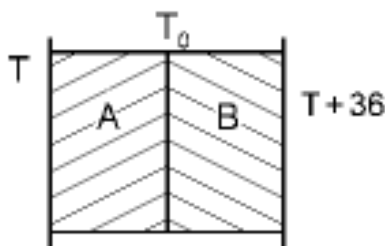
The two are in series.

$$K_A = 3K_B$$

i.e. thermal resistance of A (say R_A) is one third that of B (say R_B)

$$\text{i.e. } R_B = 3 R_A$$

$$\square \text{ Thermal current} = \frac{36}{4R_A} = \frac{9}{R_A}$$



$$\square \text{ Temperature difference across 'A'} = \left(\frac{9}{R_A} \right) \cdot R_A = 9^\circ\text{C}$$

Final Answer: Option (B) is Correct.

19)

Explain Question: It has been asked to find the dependency of frequency of photon in electron transition of higher orbit.

$$f = k \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Concept:

Solution:

$$\nu \propto \frac{1}{(n-1)^2} - \frac{1}{n^2}$$

$$\nu \propto \frac{1}{n^3}$$

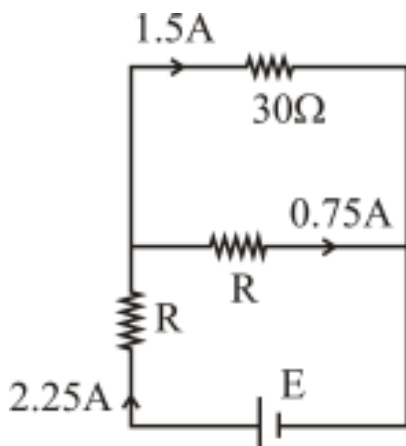
Final Answer: Option (C) is Correct

20)

Explain Question: Value of current in a branch of electric circuit has been asked.

Concept: $V = iR$ (Ohm's law)

Solution:



$$E = 2.25R + 0.75 R$$

$$E = 3R$$

$$\frac{E}{R} = 3$$

Final Answer: Option (A) is Correct.

21)

Explain Question: Magnetic field value has been asked across a ballistic galvanometer.

Concept: $q = \frac{\Delta\phi}{R}$

Solution:

$$e = iR = \frac{d\phi}{dt}$$

$$= \frac{dq}{dt} R = \frac{d\phi}{dt}$$

$$= dq R = d\phi$$

$$2 \times 10^{-5} \times 20 \times 200 = B \times 3.2 \times 10^{-4}$$

$$800 = 3.2 \times 5 \times B$$

$$B = 5$$

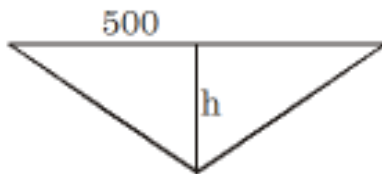
Final Answer: 5

22)

Explain Question: The expression of metal has been asked when temperature has been risen.

Concept: $\Delta l = \ell \alpha \Delta T$

Solution:



$$(500)^2 + h^2 = [500(1 + 14 \times 10^{-6} \times 5)]^2$$

$$= (500)^2 [1 + 14 \times 10^{-5}]$$

$$h^2 = (500)^2 \times 140 \times 10^{-6}$$

$$= 25 \times 10^{-1} \times 14$$

$$h^2 = 35$$

$$h = 6\text{m}$$

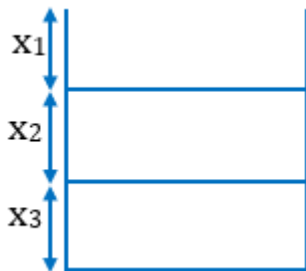
Final Answer: 6

23)

Explain Question: Apparent depth of the bucket filled of multiple liquid has been asked.

Concept: $\frac{d_{app}}{n_{eq}} = \sum \frac{t_i}{n_i}$

Solution:



$$x_1 + x_2 + x_3 = 60\text{cm} \quad \dots(i)$$

$$x_1 = \frac{x_2}{2} = \frac{x_3}{1.5} \quad \dots(ii)$$

$$\text{Apparent depth} = x_1 + \frac{x_2}{2} + \frac{x_3}{1.5}$$

Final Answer: 8

24)

Explain Question: The intensity of em wave has been asked in the question.

Concept: $I = \frac{1}{2} \epsilon_0 E_0^2$

Solution:

Average intensity $I = \frac{\epsilon_0 C E_0^2}{2}$
 $\epsilon_0 = 8.85 \times 10^{-12}$
 $C = 3 \times 10^8$
 $E_{\text{rms}} = 36$
 $I = \frac{8.85 \times 3 \times 36 \times 36 \times 10^{-4}}{2} = 1.72 \frac{\text{W}}{\text{m}^2}$
 $\Rightarrow \sqrt{3}$

Final Answer: 3

25)

Explain Question: Length of the wire has been asked in stretched string, fixed at both ends

Concept: $f = \frac{nv}{2L}$

Solution:

$$6.28 = \frac{2\pi}{\lambda} = 62.8$$

$$\lambda = 0.1\text{m}$$

$$\ell = 40 \frac{\lambda}{2} = 2\text{m}$$

Final Answer: 2

PART 2 : CHEMISTRY

26)

Explain Question: Calculate ΔS for the given reaction

Concept: $\Delta S_{\text{reaction}} = \sum \text{Entropy of product} - \sum \text{Entropy of reactant}$

Solution:

$$\Delta S_{\text{Reaction}} = [186.264] - [5.740 + 2 \times 130.684]$$

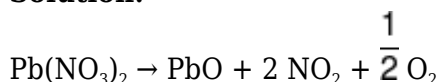
$$= -80.844 \text{ J/K}$$

Final Answer: (A)

27) **Explain Question:** Identify gas which is present in brown fumes.

Concept: Characteristics of NO_2 & O_2

Solution:



The yellow-brown fumes are characteristic of nitrogen dioxide. When a glowing splinter is held in fumes, is relighted. This indicates presence of oxygen gas. Thus, the fumes consists of NO_2 and O_2 .

Final Answer: (D)

28)

Explain Question: Identify the correct structure from given compounds.

Concept: Apply VSEPR theory

Solution:

- (A) θ incorrect less than $109^\circ 28'$
 (B) incorrect position of lone pair
 (C) correct square planar structure with $\theta = 90^\circ$
 (D) θ incorrect because it is square planar with two lone pair at Xe and $\theta = 90^\circ$

Final Answer: (C)

29) **Explain Question:** Calculate the potential difference by which a proton is accelerated.

Concept: Apply concept of de Broglie's wavelength.

Solution:

$$\lambda = \frac{h}{\sqrt{2m_p eV}} \quad \lambda_\alpha = \frac{h}{\sqrt{2m_\alpha (2e)V_\alpha}}$$

$$= 2m_p eV = 2 \times 4 \times m_p \times 2eV_\alpha$$

$$\Rightarrow V_\alpha = \frac{V}{8}$$

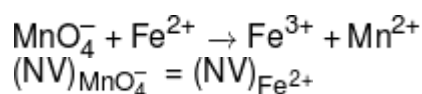
Final Answer: (D)

30)

Explain Question: Calculate the molarity of gives FeSO_4 solution

Concept: Apply the concept of equivalence

Solution:



$$32 \times 5 \times 5 = 250 \times M \times 1 \Rightarrow M = 3.2$$

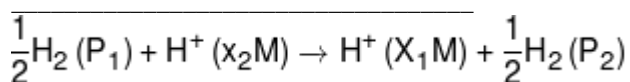
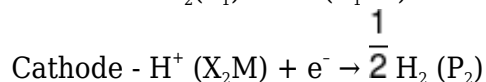
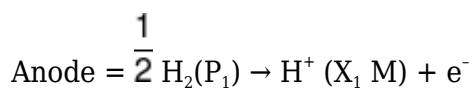
Final Answer: (A)

31)

Explain Question: Identify correct condition in which cell reaction must be spontaneous.

Concept: Cell representation & nernst equation

Solution:



$$E_{\text{cell}} = 0 - \frac{0.0591}{1} \log \frac{[\text{X}_1][\text{P}_2]^{\frac{1}{2}}}{[\text{X}_2][\text{P}_1]^{\frac{1}{2}}}$$

if $\text{P}_1 = \text{P}_2$ then for spontaneous reaction $\text{X}_2 > \text{X}_1$

if $\text{X}_1 = \text{X}_2$ then for spontaneous reaction $\text{P}_1 > \text{P}_2$

Final Answer: (C)

32)

Explain Question: Calculate activation energy in the absence of catalyst

Concept: $\ln \frac{K}{K'} = \frac{E_a - E_p}{RT}$

Solution:

$$\ln \frac{K}{K'} = \frac{E_a - E_p}{RT}$$

$$\ln 2.718 = \frac{E_a - 4.15 \times 10^3}{8.3 \times 500}$$

$$E_a = 4.15 \times 10^3 + 4.15 \times 10^3$$

$$E_a = 8.30 \text{ kJ}$$

Final Answer: (D)

33)

Explain Question: Identify correct set of properties for given complexes

Concept: Concept of hybridisation & weak and strong field ligands

Solution:

$[\text{Ni}(\text{CN})_4]^{2-}$ - dsp^2 , square planar, $\mu = 0$

$[\text{MnBr}_4]^{2-}$ - sp^3 , tetrahedral $\mu = \sqrt{35}$

$[\text{FeF}_6]^{4-}$ - sp^3d^2 , octahedral, $\mu = \sqrt{24}$

Final Answer: (D)

34)

Explain Question: Calculate potential energy of electron present in 2nd orbit of Li^{+2}

Concept: $\text{PE} = 2\text{TE}$

$$= 2 \left[-\frac{1}{2} \frac{kze^2}{r_n} \right]$$

Solution:

$$\text{PE} = 2\text{TE}$$

$$= 2 \left[-\frac{1}{2} \frac{kze^2}{r_n} \right]$$

$$r_n = 0.529 \frac{n^2}{Z} \text{Å} = r_0 \frac{n^2}{Z} \text{Å}$$

$$\text{P.E.} = -\frac{1 \times 3 \times e^2}{4\pi\epsilon_0 \times \frac{4r_0}{3}}$$

$$\text{P.E.} = -\frac{9e^2}{16\pi\epsilon_0 r_0}$$

Final Answer: (D)

35)

Explain Question: Identify true & false statement from given statements

Concept: Apply concept of ionisation energy & electron affinity

Solution:

(I) Ionisation Energy

s- electron > p- electron

(Due to penetration power)

(II) EA order $\text{Ne} < \text{Be} < \text{N} < \text{B} < \text{Li} < \text{C} < \text{O} < \text{F}$

(III) IE $\text{Mg} > \text{Al}$

$(3s^2) \quad (3p^1)$

(IV) Bond Energy order

$\text{Cl}_2 > \text{F}_2$

Final Answer: (A)

36) **Explain Question:** Identify correct statements

Concept: Apply concept of MOT

Solution:

(A) B.O. of $\text{N}_2^- = \frac{10-5}{2} = 2.5$

(B) In C_2 Bonding e^- Present in π_{2p_x} & π_{2p_y} orbitals.

(C) B.O. of $\text{C}_2^{2-} = 3$

Final Answer: (D)

37)

Explain Question: Calculate change in internal energy

Concept: Relation between ΔH (enthalpy change) and ΔE (internal energy change) is

$$\Delta H = \Delta E + \Delta n_g RT$$

Solution:

Relation between ΔH (enthalpy change) and ΔE (internal energy change) is

$$\Delta H = \Delta E + \Delta n_g RT$$

Where, Δn_g = (moles of gaseous products) - (moles of gaseous reactants)

For the given reaction, $\Delta n_g = 2 - 3 = -1$

$$\Rightarrow -1366.5 = \Delta E - 1 \times 8.314 \times 10^{-3} \times 300$$

$$\therefore \Delta E = -1364.0 \text{ kJ mol}^{-1}$$

Final Answer: (C)

38)

Explain Question: Identify correct order of thermal stability of given carbonates

Concept: Thermal stability $\propto$ size of cation

Solution:

For poly atomic anion

Thermal stability $\propto$ size of cation

Order of size of Cation - $\text{Be}^{2+} < \text{Mg}^{2+} < \text{Ca}^{2+} < \text{K}^+$

Thermal stability order $\text{BeCO}_3 < \text{MgCO}_3 < \text{CaCO}_3 < \text{K}_2\text{CO}_3$

Final Answer: (B)

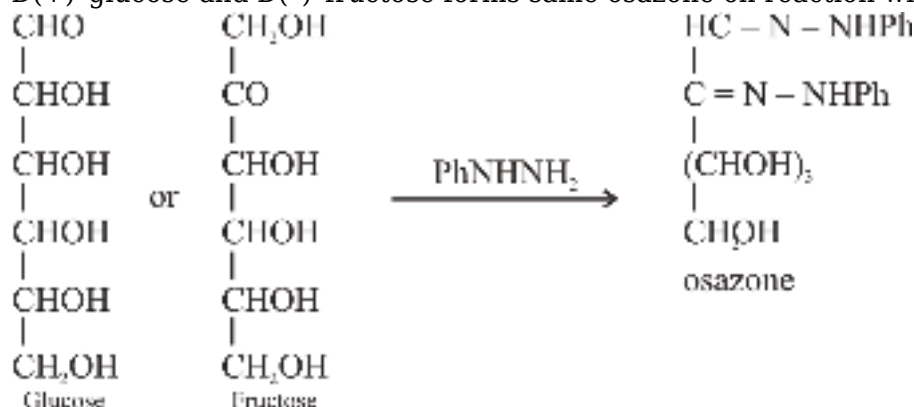
39)

Explain Question: Identify correct statement

Concept: D(+)-glucose and D(-)-fructose forms same osazone on reaction with PhNHNH_2

Solution:

D(+)-glucose and D(-)-fructose forms same osazone on reaction with PhNHNH_2



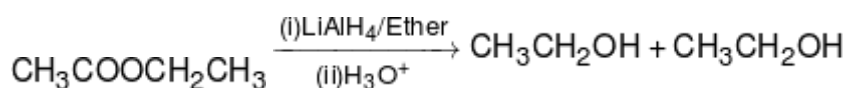
Final Answer: (B)

40)

Explain Question: Identify reagent which used for given conversion

Concept: Reduction of esters to alcohols

Solution:

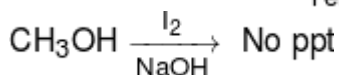
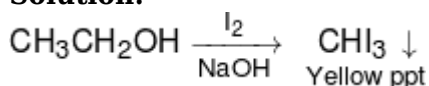


Final Answer: (B)

41) **Explain Question:** Identify reagent which used to distinguish methyl and ethyl alcohol.

Concept: Iodoform reaction & mechanism

Solution:



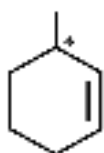
Final Answer: (C)

42)

Explain Question: Identify compound which exhibit optical isomerism

Concept: Concept of optical isomerism

Solution:



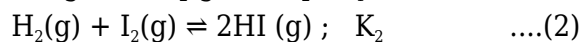
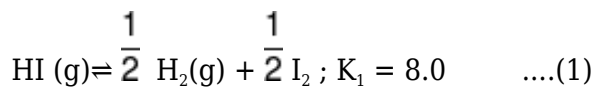
Compound has non superimposable mirror image.

Final Answer: (C)

43) **Explain Question:** Calculate the equilibrium constant for given reaction

Concept: Apply the concept of equilibrium constant

Solution:



$$K_2 = \frac{1}{K_1^2}$$

$$K_2 = \frac{1}{8^2} = \frac{1}{64}$$

Final Answer: (A)

44) **Explain Question:** Identify substituent which favours the hoffmann elimination

Concept: hoffmann alkene product will be formed when X is poor leaving group

Solution:

In the above case hoffmann alkene product will be formed when X is poor leaving group. So, here among halogens fluorine is poor leaving group.

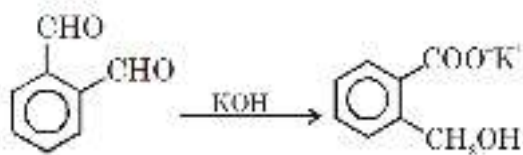
Final Answer: (D)

45) **Explain Question:** Identify product 'B' in the given reaction

Concept: Intramolecular Cannizzaro reaction

Solution:

Given reaction is an example of intramolecular Cannizzaro reaction.



Final Answer: (B)

46) **Explain Question:** Identify elements which gives flame test.

Concept: High I.E. elements do not gives flame test

Solution:

Be & Mg do not give flame test due to high I.E.

Final Answer: (6)

47) **Explain Question:** Identify compounds which formed black ppt.

Concept: Colours of sulphides of different elements

Solution:

Ag_2S - Black

PbS - Black

HgS - Black

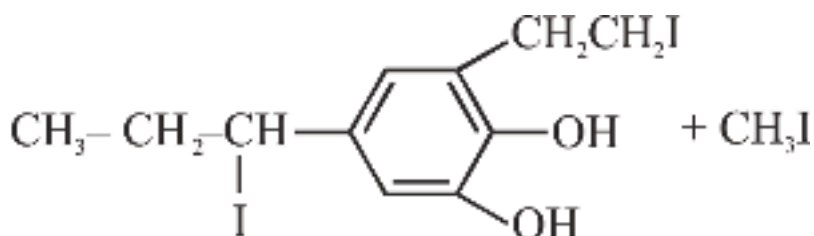
CuS - Black

Final Answer: (4)

48) **Explain Question:** Find total moles of HI used for given compounds.

Concept: Mechanism of iodoform and addition reaction

Solution:



Final Answer: (3)

49) **Explain Question:** Identify total number of chiral molecules

Concept: Apply concept of chirality

Solution:

Option (B), (C), (D) and (E) are chiral due to absence of plane of symmetry, centre of symmetry and alternate axis of symmetry.

Final Answer is = 4

Final Answer: (4)

50) **Explain Question:** Calculate no. of moles of gas dissolved

Concept: Apply concept of Henry's law

Solution:

$$P_T = P_x + P_{\text{H}_2\text{O}}$$

$$P_x = 1.5 - 0.5 = 1 \text{ atm}$$

$$\text{slope of graph} = K_H = 5 \times 10^3 \text{ atm}$$

$$P_x = K_H \frac{n_x}{n_{\text{H}_2\text{O}}}$$

$$(\because n_x \ll n_{\text{H}_2\text{O}} \text{ so } n_x + n_{\text{H}_2\text{O}} \approx n_{\text{H}_2\text{O}})$$

$$1 = 5 \times 10^3 \times \frac{n_x}{10}$$

$$n_x = \frac{1}{500} = 2 \times 10^{-3}$$

$$x = 2$$

Final Answer: (2)

PART 3 : MATHEMATICS

51) **Question Explanation:**

In this question $a + b + c = 0$ is given and based on this condition. We need to determine the nature of a complex algebraic expression involving a, b, c raised to powers up to 6 :

$$a^6 + b^6 + c^6 - \frac{(a^2 + b^2 + c^2)^3}{4} - \frac{(a^3 + b^3 + c^3)^2}{3}$$
 and we have to check the value of this expression is always positive negative or independent of a, b, c .

Concept:

This question is based on Theory of equation & Newton's formula.

Solution:

$$\text{Given } a + b + c = 0$$

$$\text{Let } p = ab + bc + ca \text{ and } q = abc$$

$$\text{Let } S_k = a^k + b^k + c^k$$

$$S_1 = a + b + c = 0$$

$$\begin{aligned} S_2 &= a^2 + b^2 + c^2 \\ &= (a + b + c)^2 - 2(ab + bc + ca) \\ &= (0)^2 - 2p \end{aligned}$$

$$S_2 = -2p$$

$$\begin{aligned} S_3 &= a^3 + b^3 + c^3 \quad (\text{as } a + b + c = 0) \\ &= 3abc \end{aligned}$$

$$S_3 = 3q$$

We can assume a cubic equation whose roots are a, b, c as-

$$\boxed{x^3 + px - q = 0}$$

use Newton's formula.

$$S_{k+3} + pS_{k+1} - qS_k = 0$$

$$S_{k+3} = qS_k - pS_{k+1}$$

$$\mathbf{k = 1}$$

$$\begin{aligned} S_4 &= qS_1 - pS_2 \\ &= q \times 0 - p(-2p) \end{aligned}$$

$$S_4 = 2p^2$$

$$\mathbf{k = 2}$$

$$\begin{aligned} S_5 &= qS_2 - pS_3 \\ &= q \times (-2p) - p(3q) \end{aligned}$$

$$S_5 = -5pq$$

$$\mathbf{k=3}$$

$$\begin{aligned} S_6 &= qS_3 - pS_4 \\ &= q \times 3q - p(2p^2) \end{aligned}$$

$$S_6 = 3q^2 - 2p^3$$

The expression to evaluate

$$\text{Is } E = S_6 - \frac{(S_2)^3}{4} - \frac{(S_3)^2}{3}$$

Put the value of S_2 , S_3 , S_6 in terms of p and q

$$\begin{aligned} &= 3q^2 - 2p^3 - \frac{(-2p)^3}{4} - \frac{(3q)^2}{3} \\ &= 3q^2 - 2p^3 + 2p^3 - 3q^2 \end{aligned}$$

$$\boxed{E = 0}$$

Since the value of this expression is '0'.

So, It is independent of the values of a, b, c.

Final Answer:

The correct option is (3)

52) **Explain Question:** To find maxima and minima for given function.

Concept: Maxima - minima.

Solution:

$$f(x) = |2x^2 + 3x - 2| + \sin x \cos x$$

$$f(x) = |(2x - 1)(x + 2)| + \sin x \cos x$$

$$f' = \begin{cases} 4x + 3 + \frac{\cos 2x}{4}, & \frac{1}{2} < x < 1 \\ -(4x + 3) + \frac{\cos 2x}{4}, & 0 \leq x < \frac{1}{2} \end{cases}$$

for $0 \leq x < \frac{1}{2} \Rightarrow f'(x) < 0$

for $\frac{1}{2} < x \leq 1 \Rightarrow f'(x) > 0$

$f(x)$ local minima at $x = \frac{1}{2}$ and

local maxima at $x = 1$

$$f\left(\frac{1}{2}\right) + f(1) = 3 + \frac{1}{2}(1 + 2 \cos 1) \sin 1$$

Final Answer: (B)

$$53) f(x) = \frac{x}{1 + |x|}$$

$$\Rightarrow f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} \frac{1}{(1-x)^2}, & x < 0 \\ \frac{1}{(1+x)^2}, & x > 0 \end{cases}$$

Here we observe that

$$\text{LHD} = 1 = \text{Lf}'(0)$$

$$\text{RHD} = 1 = \text{Rf}'(0)$$

□ Function is differentiable over $(-\infty, \infty)$

54) **Explain Question:** To solve trigonometric equation.

Concept: Trigonometric equation.

Solution:

Let $3^x \pi = \theta$ then

$$\begin{aligned}
 &\Rightarrow \cos \theta - 2 \cos^2 \theta + 2 \cos 4\theta - \cos 7\theta \\
 &= \sin \theta + 2 \sin^2 \theta - 2 \sin 4\theta + 2 \sin 3\theta - \sin 7\theta \\
 &\Rightarrow \cos \theta - \cos 7\theta + \sin 7\theta - \sin \theta \\
 &\quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \\
 &\Rightarrow 2 \sin 4\theta \sin 3\theta + 2 \cos 4\theta \sin 3\theta \\
 &\quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \\
 &\Rightarrow \sin 4\theta (\sin 3\theta + 1) + \cos 4\theta (\sin 3\theta + 1) \\
 &\quad - (1 + \sin 3\theta) = 0 \\
 &\Rightarrow (\sin 3\theta + 1)(\sin 4\theta + \cos 4\theta - 1) = 0 \\
 &\Rightarrow x = \log_3 \frac{4n-1}{2} - 1 \text{ And } \sin 4\theta + \cos 4\theta = 1 \\
 &\Rightarrow \sin \left(4\theta + \frac{\pi}{4} \right) = \sin \frac{\pi}{4} \Rightarrow 4\theta + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4} \\
 &\Rightarrow x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16} \right), n \in \mathbb{Z}
 \end{aligned}$$

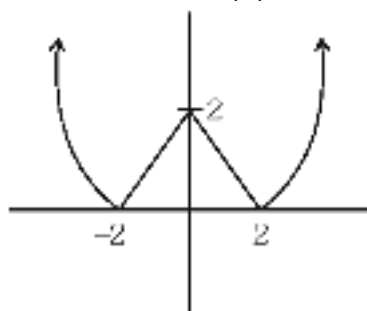
Final Answer: (A)

55) **Explain Question:** To find number of points where function is non-differentiable and number of real solution.

Concept: Differentiability.

Solution:

$$\begin{aligned}
 f(x) &= \frac{(x^2 - 4) + (2 - |x|)}{2} + \left| \frac{(x^2 - 4) - (2 - |x|)}{2} \right| \\
 &= \max. \{x^2 - 4, 2 - |x|\}
 \end{aligned}$$



$$\Rightarrow \ell = 3 \text{ \& } m = 3$$

$$\Rightarrow (\ell + m) = 6$$

Final Answer: (C)

56) **Explain Question:** To find value of $a_1 + b_1$ for given condition.

Concept: Complex numbers.

Solution:

The point $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n), (a_{n+1}, b_{n+1})$

(where x is $\overline{2 + \sqrt{3}} = 2 - \sqrt{3}$)

$$\cos(\pi - 2\theta) = \frac{\frac{x^2}{4} + \frac{x^2}{4} - x'^2}{2 \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)}$$

$$\frac{-x^2}{2} \cos 2\theta = \frac{x^2}{2} - x'^2$$

$$x'^2 = \frac{x^2}{2} (1 + \cos 2\theta) = x^2 \cos^2 \theta$$

$$\square \quad x' = x \cos \theta.$$

Final Answer: (D)

58) **Explain Question:** To solve differentiable equation.

Concept: Linear differentiable equation.

Solution:

$$e^x \left(\int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 \right) = 1 \int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 = e^{-x}$$

$$\int_0^x f(t) dt + \int_0^x (x-t) f(t) dt + 1 = e^{-x}$$

$$\int_0^x f(t) dt + x \int_0^x f(t) dt - \int_0^x t f(t) dt + 1 = e^{-x} \quad \dots(1)$$

$$f(x) + xf(x) + \int_0^x f(t) dt \cdot 1 - xf(x) = -e^{-x}$$

$$(x+1)f(x) + \int_0^x f(t) dt = xf(x) - e^{-x} \quad \dots(1)$$

$$(x+1)f'(x) + f(x) + f(x) = xf'(x) + f(x) + e^{-x}$$

$$f'(x) + f(x) = e^{-x}$$

$$(e^x f(x))' = 1 \Rightarrow e^x f(x) = x + c$$

$$\text{in (1) put } x = 0 \quad f(0) = -1$$

$$\therefore f(0) = c \Rightarrow c = -1$$

$$e^x f(x) = x - 1 \Rightarrow f(x) = \frac{x-1}{e^x}$$

$$f(2) = \frac{1}{e^2}$$

Final Answer: (B)

59) **Explain Question:** To find area of triangle formed by tangent at $x = \frac{1}{e}$ to curve $y = f(x) + 1$ and co-ordinate axes.

Concept: Area under curve.

Solution:

$$f(x) = 6 (\ln x)^2 - 1$$

$$\int_1^e \frac{f(t)}{t} dt$$

$$f(x) = 6 (\ln x)^2 - \alpha \quad \dots(1) \quad \left(\int_1^e \frac{f(t)}{t} dt = \alpha \right)$$

$$\begin{aligned} \text{Now, } \alpha &= \int_1^e \frac{f(t)}{t} dt \\ &= \int_1^e \frac{(6(\ln t)^2 - \alpha)}{t} dt \\ &= 6 \int_1^e \frac{(\ln t)^2}{t} dt - \alpha \cdot \int_1^e \frac{1}{t} dt \\ &= 6 \cdot \left(\frac{(\ln t)^3}{3} \right)_1^e - \alpha \cdot (\ln t)_1^e \\ &= 2 \cdot (1^3 - 0^3) - \alpha(1 - 0) \\ &= 2 - \alpha \\ &\Rightarrow \alpha = 1 \end{aligned}$$

$$\text{So, } f(x) = 6 (\ln x)^2 - 1$$

$$\text{Let } y = g(x) = f(x) + 1 = 6 (\ln x)^2$$

$$g'(x) = 6 \cdot 2 \cdot (\ln x) \cdot \frac{1}{x} = 12 \cdot \left(\frac{\ln x}{x} \right)$$

$$g\left(\frac{1}{e}\right) = 12 \cdot \frac{\ln\left(\frac{1}{e}\right)}{\frac{1}{e}} = 12e \cdot (-1) = -12e$$

$$g'\left(\frac{1}{e}\right) = 6 \cdot \ln\left(\frac{1}{e}\right)^2 = 6 \cdot (-\ln e)^2 = 6$$

$$P\left(\frac{1}{e}, 6\right)$$

$$T: y - 6 = -12e \left(x - \frac{1}{e} \right)$$

$$T: y - 6 = -12ex + 12$$

$$y + 12ex = 18$$

$$x = 0 \Rightarrow y = 18$$

$$y = 0 \Rightarrow x = \frac{3}{2e}$$

$$\Delta = \frac{1}{2} \times 18 \times \frac{3}{2e} = \frac{27}{2e}$$

$$\frac{2e\Delta}{3} = 9$$

$$\left[\frac{2e\Delta}{3} \right] = 9$$

Final Answer: (D)

60) **Explain Question:** Indefinite integration.

Concept: indefinite integration.

Solution:

$$\int \frac{(e^{2x} - e^x + 1) dx}{(e^x \sin x + \cos x)(e^x \cos x - \sin x)}$$

Given $f(0) = 2025$

$$\rightarrow f(x) = e^x \sin x + \cos x$$

$$f'(x) = e^x \cos x + (\sin x)e^x - \sin x$$

$$g(x) = e^x \cos x - \sin x$$

$$g'(x) = \cos x \cdot e^x - e^x \sin x - \cos x$$

Now

$$f(x) \cdot g'(x) = (e^x \sin x + \cos x)(\cos x \cdot e^x - e^x \sin x - \cos x)$$

$$= e^{2x} \sin x \cos x - e^{2x} \sin^2 x - e^x \sin x \cos x + e^x \cos^2 x$$

And

$$g(x) \cdot f'(x) = (e^x \cos x - \sin x)(e^x \cos x + \sin x \cdot e^x - \sin x)$$

$$= e^{2x} \cos^2 x + e^{2x} \sin x \cos x - e^x \sin x \cos x - e^x \sin x \cos x$$

$$- e^x \sin^2 x + \sin^2 x$$

$$I = \int \frac{f(x) \cdot g'(x) - g(x) \cdot f'(x)}{f(x) g(x)} dx = \ln \left| \frac{f(x)}{g(x)} \right| + C$$

$$f(x) = I = \ln \left| \frac{e^x \cos x - \sin x}{(e^x \sin x + \cos x)} \right| + C$$

$$f(0) = \ln \left| \frac{1}{1} \right| + C = \pm 2025$$

$$C = 2025$$

$$f(2024\pi) = \ln \left| \frac{e^{2024} \cdot 1 - 0}{e^{2024} \cdot 0 + 1} \right| + 2025$$

$$= 2024\pi \pm 2025$$

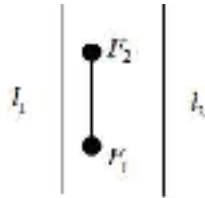
Option B correct

Final Answer: (A)

61) **Explain Question:** Two parabolas intersect at distinct points A and B, to find AB^2 .

Concept: Parabola.

Solution:



$$P_1 : y^2 = 4ax \quad P_2 : (y-b)^2 = -4a(x-a)$$

$$P_1(a, 0) P_2(h-b, b)$$

$$P_1 P_2 \text{ parallel to } l_1, l_2 \Rightarrow h-b=a$$

$$\Rightarrow h=a+b$$

$$P_1 \text{ lies on } P_2 \Rightarrow b^2 = -4a(a-b) = -4ab^2 \quad (1)$$

$$P_2 \text{ lies on } P_1 \Rightarrow b^2 = 4a(h-b) = 4a^2 \quad (2)$$

$$a=b \quad P_1(a, 0) P_2(a, -2a) \Rightarrow P_1 P_2 = 2a$$

$$A \text{ and } B \text{ are points of intersections of } y^2 = 4ax \text{ and } (y-2a)^2 = -4a(x-2a)$$

$$(y-2a)^2 = 4ax + 8a^2 \quad y^2 = 8a^2 + y^2 = 4ay + 4a^2 \quad y^2 = 8a^2$$

$$\Rightarrow 2y^2 - 4ay - 4a^2 = 0 \Rightarrow y^2 - 2ay - 2a^2 = 0$$

$$y_1 + y_2 = 2a, y_1 y_2 = -2a^2, y_1 - y_2 = \sqrt{12a^2}, AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$x_2 - x_1 = \frac{y_2^2}{4a} - \frac{y_1^2}{4a} = \frac{(y_2 - y_1)(y_1 + y_2)}{4a} = \frac{y_1 - y_2}{2}$$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{\left(\frac{y_1 - y_2}{2}\right)^2 + (y_1 - y_2)^2} = \frac{y_1 - y_2 \sqrt{5}}{2}$$

$$= \frac{(2\sqrt{3}a)(\sqrt{5})}{2} = \sqrt{15}a \quad \frac{\sqrt{15}}{2} \quad AB = \frac{15}{4} = \frac{m}{n}$$

Final Answer: (D)

62) **Explain Question:** Inverse and composite function.

Concept: Inverse and composite function.

Solution:

$$f \circ g \circ g^{-1} = g^{-1} \circ g^{-1} \text{ (Apply } g^{-1} \text{ on both side)}$$

$$\Rightarrow f = g^{-1} \circ g^{-1} \text{ (since } g \circ g^{-1} \text{ is identity)}$$

$$\text{For } g(x) = (x+3)^3 + 54$$

$$g^{-1}(x) = (x-54)^{1/3} - 3$$

$$g^{-1}(-289) = -10 \text{ and } g^{-1}(-10) = -7$$

$$|f(-289)| = 7$$

Final Answer: (C)

63) **Explain Question:** Hyperbola and ellipse.

Concept: Hyperbola and ellipse.

Solution:

$$H : \frac{x^2}{3} - y^2 = 1, \text{ Let } E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\square E \text{ and } H \text{ are confocal} \Rightarrow a^2 = \frac{16}{3}, b^2 = \frac{4}{3}$$

$$\Rightarrow E: \frac{3x^2}{16} + \frac{3y^2}{4} = 1$$

$$P\left(2, \frac{1}{\sqrt{3}}\right)$$

Solving both curves

equation of tangent to E and P is

$$\frac{3 \cdot x \cdot 2}{16} + \frac{3 \cdot y \cdot 1}{4 \cdot \sqrt{3}} = 1$$

$$\text{equation of tangent to H at P is } \frac{2x}{3} - \frac{y \cdot 1}{\sqrt{3}} = 1$$

$$\Rightarrow N_1\left(\frac{8}{3}, 0\right) \text{ and } T_1\left(\frac{3}{2}, 0\right)$$

$\square$ triangle T_1PN_1 is right angled

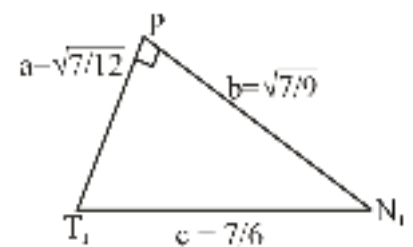
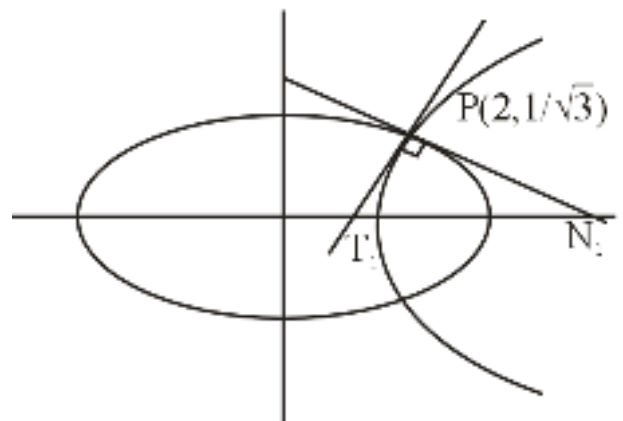
$$\square r = \frac{ab}{a+b+c}$$

$$= \frac{\frac{7}{\sqrt{108}}}{\sqrt{\frac{7}{12}} + \sqrt{\frac{7}{9}} + \frac{7}{6}}$$

$$= \frac{\frac{1}{\sqrt{84}} + \frac{1}{\sqrt{63}} + \frac{1}{\sqrt{36}}}{\frac{1}{\sqrt{108}}}$$

$$= \frac{1}{\sqrt{84}} + \frac{1}{\sqrt{63}} + \frac{1}{\sqrt{36}}$$

Final Answer: (C)



64) **Explain Question:** Ellipse and Circle.

Concept: Ellipse and Circle.

Solution:

$$Q(-4 \cos \theta, 3 \sin \theta)$$

$$R(4 \cos \theta, 3 \sin \theta)$$

$$\text{Slope of PQ} = \frac{3 \sin \theta + 3}{-4 \cos \theta - 3 \sin \theta + 3}$$

$$\text{Slope of PR} = \frac{4 \cos \theta}{3 \sin \theta + 3}$$

$PQ \perp PR$

$$\frac{3(\sin \theta + 1) \cdot 3(\sin \theta + 1)}{-16 \cos^2 \theta} = -1$$

$$9(\sin^2 \theta + 2 \sin \theta + 1) = 16(1 - \sin^2 \theta)$$

$$25 \sin^2 \theta + 18 \sin \theta - 7 = 0$$

$$\sin \theta = \frac{7}{25}, \sin \theta = -1 \text{ (Reject)}$$

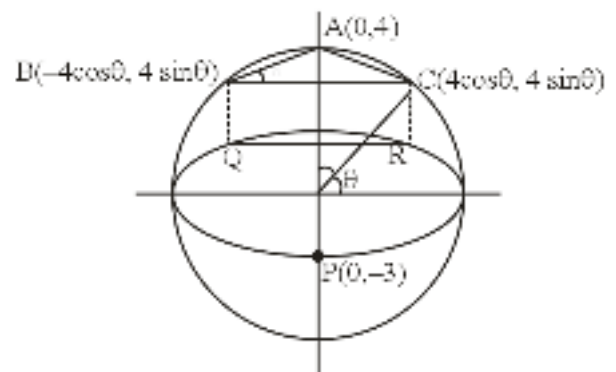
$$\text{Let } \phi \text{ is smallest angle} = \angle ABC = \frac{1}{2} \left(\frac{\pi}{2} - \theta \right)$$

$$\phi = \frac{\pi}{4} - \frac{\theta}{2}$$

$$\tan \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{1}{7}$$

$$\tan \phi = \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}} = \frac{3}{4}$$

Final Answer: (A)



65) **Explain Question:** Inverse trigonometric function and vectors.

Concept: Inverse trigonometric function and vectors.

Solution:

We have

$\tan^{-1} x + \cot^{-1} x = \sin^{-1} x + \cos^{-1} x$, which is true for all $x \in [-1, 1]$

So, number of integral values of $x = 3$.

$$\vec{c} = \left| \vec{a} \times \vec{b} \right| \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b}, |\vec{a}| = 1 = \left| \vec{b} \right| \vec{a} \wedge \vec{b} = \frac{\pi}{6}$$

$$\square |\vec{a} \times \vec{b}| = (1)(1) \sin \frac{\pi}{6} = \frac{1}{2} \quad \vec{a} \cdot \vec{b} = (1)(1) \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \vec{c} = \frac{1}{2} \vec{a} + \frac{\sqrt{3}}{2} \vec{b}$$

$$\Rightarrow |2\vec{c}|^2 = \left| \vec{a} + \sqrt{3}\vec{b} \right|^2$$

$$\Rightarrow 4|\vec{c}|^2 = 1 + 3 + 2\sqrt{3} \left(\frac{\sqrt{3}}{2} \right) = 7$$

$$\square 4|\vec{c}|^2 = 7$$

Final Answer: (C)

66) **Explain Question:** Limits.

Concept: Limits.

Solution:

$$\text{Slope of normal} = -\frac{1}{f'(0)} = 3 \Rightarrow f'(0) = -\frac{1}{3}$$

$$\lim_{x \rightarrow 0} \frac{x^2}{\{f(x^2) - 5f(4x^2) + 4f(7x^2)\}}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{\frac{1}{f(x) - 5f(4x) + 4f(7x)}}$$

$$= \frac{f'(0) - 20f'(0) + 28f'(0)}{1}$$

$$= \frac{1}{9f'(0)} = \frac{1}{9} \times -3 = -\frac{1}{3}$$

Final Answer: (C)

67) **Explain Question:** Rate measure.

Concept: Rate measure.

Solution:

Distance covered by X after time $t = 4t$

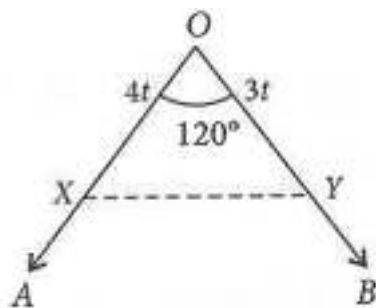
Similarly by Y after time $t = 3t$

Let shortest distance between X and Y = A

$$\Rightarrow A^2 = 16t^2 + 9t^2 - 24t^2 \cos 120^\circ \text{ (By cosine law)}$$

$$\Rightarrow A^2 = 25t^2 - 24t^2 \left(\frac{-1}{2} \right) = 25t^2 + 12t^2 = 37t^2$$

$\therefore$ A after $t = 1$ hr is $\sqrt{37}$ km



Now differentiating w.r.t t we get

$$2AA' = 2 \cdot 37t. \text{ After } t = 1 \text{ hr}$$

$$2 \cdot \sqrt{37}A' = 2 \cdot 37 \Rightarrow A' = \sqrt{37}$$

$\Rightarrow$ Rate at which shortest distance A changes with time = $\sqrt{37}$ km/hr.

Final Answer: (A)

68) **Explain Question:** Limits for composite function.

Concept: Limits for composite function.

Solution:

$$(A) \lim_{x \rightarrow 1^+} g(f(x)) = \lim_{h \rightarrow 0} g(|-1+h| - 1 - h) = g(0^-) = \text{does not exist}$$

$$(B) \lim_{x \rightarrow 0^-} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) \\ = \lim_{h \rightarrow 0} [2-h] - \cos(2-h-2) = 1 - 1 = 0$$

$$(C) \lim_{x \rightarrow 2^+} \frac{f(g(x))}{f(x) - 2} = \lim_{h \rightarrow 0} \frac{f(g(2+h))}{f(2+h) - 2} \\ = \lim_{h \rightarrow 0} \frac{f(2 - \cos h)}{h - (2+h) - 2} = \lim_{h \rightarrow 0} \frac{f(2 - \cos h)}{-4} \\ = \lim_{x \rightarrow 0} \frac{|2 - \cos h - 2| - (2 - \cos h)}{-4} \\ = 1 - 1 = 0$$

$$(D) = \lim_{x \rightarrow 0^+} \frac{g(f(x))}{(f(x) - 2)^2} = \lim_{h \rightarrow 0} \frac{g(2-h)}{h^2} \\ = \lim_{h \rightarrow 0} \frac{1 - \cosh}{h^2} = \frac{1}{2}$$

Final Answer: (B)

69) **Question Explanation:** Determine the value of $P(E) + P(F)$, given E, F are independent and compound probabilities.

Concept: Independent events.

Solution:

(A) E & F are independent events.

$$\text{so } P(E \cap F) = P(E)P(F) \quad \text{_____ (i)}$$

$$\text{Given } P(E \cap F) = \frac{1}{12} \text{ and } P(\bar{E} \cap \bar{F}) = \frac{1}{2} P(\bar{E} \cap \bar{F}) = \frac{1}{2}$$

$$1 - P(E \cup F) = \frac{1}{2}$$

$$P(E \cup F) = \frac{1}{2} \quad \text{_____ (ii)}$$

From (ii), using Addition theorem of probability,

$$P(E \cup F) = \frac{1}{2}$$

$$\Rightarrow P(E) + P(F) - P(E \cap F) = \frac{1}{2}$$

$$\Rightarrow P(E) + P(F) - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow P(E) + P(F) = \frac{1}{2} \quad \text{(iii)}$$

$$\text{From (i)} \quad P(E)P(F) = \frac{1}{12} \quad \text{(iv)}$$

$$\text{solving (iii) and (iv), we get } P(E) = \frac{4}{12} \text{ or } \frac{3}{12}$$

$$\text{and } P(F) = \frac{3}{12} \text{ or } \frac{4}{12}$$

$$\frac{P(E)}{P(F)} \text{ can be } \frac{4}{3} \text{ or } \frac{3}{4}$$

$$\text{Final Answer: } \frac{4}{3}$$

70) **Explain Question:** Three or more vectors.

Concept: Three or more vectors.

Solution:

$$\begin{bmatrix} \vec{AB} & \vec{AC} & \vec{AD} \end{bmatrix} = 0 \forall t \in \mathbb{R}$$

$$\begin{vmatrix} 3 & f(t) & 1 \\ 2 & f'(t) & 2 \\ 1 & 0 & 0 \end{vmatrix} = 0 \forall t \in \mathbb{R}$$

If D is $\hat{i}$ then

$$\therefore f'(t) = 2f(t) \Rightarrow f(t) = C \cdot e^{2t}, C \in \mathbb{R}$$

$$\begin{vmatrix} 3 & f(t) & 1 \\ 2 & f'(t) & 2 \\ 1 & 1 & 0 \end{vmatrix} = 0 \forall t \in \mathbb{R}$$

If D is $\hat{i} + \hat{j}$ then

$$\Rightarrow 2f(t) - f'(t) - 4 = 0$$

$$\therefore f'(x) - 2f(x) = -4$$

Now, I.F. = e^{-2x}

$$\therefore y \cdot e^{-2x} = \int -4e^{-2x} dx$$

$$y \cdot e^{-2x} = 2e^{-2x} + C$$

$$\therefore y = f(x) = 2 + C \cdot e^{-2x}$$

Final Answer: (C)

71) **Explain Question:** LMVT.

Concept: LMVT.

Solution:

$$\text{By LMVT } f'(c_1) = \frac{f(0) - f(-5)}{5}$$

$$-3 \leq f'(c_1) \leq 3$$

$$g(c_1) = f(c_1) f'(c_1) \quad g(c_1) \in [-54, 54]$$

$$\text{similarly } g(c_2) \in [-54, 54]$$

$$g'(x) = (f(x))^2 + f(x) f'(x)$$

$$g''(x) = 2f'(x) f''(x) + f'(x) f''(x) + f(x) f'''(x)$$

$$g''(c) = 3f'(c) f''(c) + f(c) f'''(c)$$

$$= \frac{-3(f'(c))^3 + (f(c))^2 f'''(c)}{f(c)} < 0$$

$$g''(c) < 0 \text{ as } f(c) \text{ has range } [3, 18]$$

so there always exist some c at which g(x) attains maxima

If g(c₁), g(c₁) has maximum 54,

then in order to ensure a maxima for all function g(0) = k > 54

then g(0) = k > 54

$$\square \quad k_{\min} = 55$$

Final Answer: (55)

72) **Explain Question:** Indefinite integration.

Concept: Indefinite integration.

Solution:

$$\begin{aligned} I_{(1,10)} &= \int \cos 2x (1 - \tan^2 x)^{10} dx \\ &= \frac{\sin 2x}{2} (1 - \tan^2 x)^{10} \\ &\quad - \int \frac{\sin 2x}{2} 10 (1 - \tan^2 x)^9 (-2 \tan x \sec^2 x) dx \\ &= \frac{\sin 2x}{2} (1 - \tan^2 x)^{10} + 20 \int \tan^2 x (1 - \tan^2 x)^9 dx \\ &= \frac{\sin 2x}{2} (1 - \tan^2 x)^{10} \\ &\quad + 20 \int (1 - (1 - \tan^2 x))(1 - \tan^2 x)^9 dx \\ I_{(1,10)} &= \frac{\sin 2x}{2} (1 - \tan^2 x)^{10} + 20I_{(0,9)} - 20I_{(0,10)} + C \\ 20I_{(0,10)} &+ I_{(1,10)} \\ &= \frac{\sin 2x}{2} (1 - \tan^2 x)^{10} + 20I_{(0,9)} + C \\ 4I_{(0,10)} &+ \frac{I_{(1,10)}}{5} \\ &= \frac{\sin 2x}{10} (1 - \tan^2 x)^{10} + 4I_{(0,9)} + C \\ A &= \frac{1}{10}; \quad B = 4 \end{aligned}$$

Final Answer: (5)

73) **Explain Question:** Parabola and ellipse.

Concept: Parabola and ellipse.

Solution:

$$\text{Let Q be } (a \cos \theta, a \sqrt{1 - e^2} \sin \theta)$$

$$\text{Slope of tangent at point Q to the ellipse is } (-\sqrt{1 - e^2} \cot \theta)$$

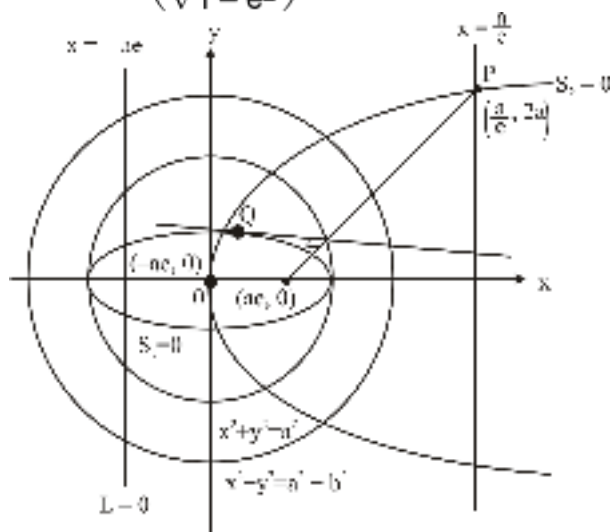
But tangent at Q will be perpendicular to the line joining P to the focus of parabola.

$$\text{So } \left(\frac{2a - 0}{\frac{a}{e} - ae} \right) \times [-\sqrt{1 - e^2} \cot \theta] = -1$$

$$\Rightarrow \frac{2e}{1-e^2} \times \sqrt{1-e^2} \cot \theta = 1$$

$$\Rightarrow \tan \theta = \frac{2e}{\sqrt{1-e^2}}$$

$$\theta = \tan^{-1} \left(\frac{2e}{\sqrt{1-e^2}} \right)$$



Final Answer: (3)

$$74) (a + b + c)^2 = 0$$

$$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ac) = 0$$

$$\Rightarrow ab + bc + ac = -\frac{1}{2}$$

again square both sides

$$(ab)^2 + (bc)^2 + (ac)^2 + 2(a^2bc + ab^2c + abc^2) = \frac{1}{4}$$

$$\Rightarrow (ab)^2 + (bc)^2 + (ac)^2 + 2abc(a + b + c) = \frac{1}{4}$$

$$\Rightarrow (ab)^2 + (bc)^2 + (ac)^2 + 2abc(0) = \frac{1}{4}$$

$$\Rightarrow (ab)^2 + (bc)^2 + (ac)^2 = \frac{1}{4} = \frac{p}{q}$$

$$p + q = 5$$

75)

Explain Question: To find minimum value of n for given.

Concept: Permutation and combination.

Solution:

First we prove $n \geq 16$. In fact, let $A_0 = \{1, 2^2, 3^2, 5^2, \dots, 41^2, 43^2\}$, where the members in A_0 , other than 1, are the squares of prime numbers not greater than 43. Then $A_0 \subseteq S$, $|A_0| = 15$ and the numbers in A_0 are pairwise coprime but A_0 contains no prime number.

Thus $n \geq 16$

Next we show that for arbitrary $A \subseteq S$ with $n = |A| = 16$, if the numbers in A are pairwise coprime, then A must contain a prime number.

In fact, if A contains no prime number, denote $A = \{a_1, a_2, \dots, a_{16}; a_1 < a_2 < \dots < a_{16}\}$. Then there are two possibilities.

(1) If $1 \notin A$, then $a_1, a_2, \dots, a_{16}$ are composite number. Since $(a_i, a_j) = 1 (1 \leq i < j \leq 16)$, the

prime factors of a_i and a_j are pairwise distinct. Let p_i be the smallest prime factor of a_i . We may assume that $p_1 < p_2 < \dots < p_{16}$, then

$$a_1 \geq p_1^2 \geq 2^2, a_2 \geq p_2^2 \geq 3^2, \dots, a_{15} \geq p_{15}^2 \geq 47^2 > 2005,$$

it leads to a contradiction.

(2) If $1 \in A$, let $a_{16} = 1$, $a_1, a_2, \dots, a_{15}$ are composite numbers. By the same assumption and argument of (1), we have $a_1 \geq p_1^2 \geq 2^2, a_2 \geq p_2^2 \geq 3^2, \dots, a_{15} \geq p_{15}^2 \geq 47^2 > 2005$. Again, it leads to a contradiction.

From (1) and (2), A contains at least one prime number, i.e. when $n = |A| = 16$, then conclusion is true. Thus, the minimum number of n is 16.

Final Answer: (16)